

CSC 302: MULTIMEDIA SIGNAL PROCESSING AND COMMUNICATIONS 3 Credits

Chapter One: Introduction to multimedia signals

Chapter Two: Multimedia Signal Representation and processing

Chapter Three: Digital Satellite Communications

Chapter Four: Multiple Access Techniques in Communication

Chapter Five: Television

CHAPTER ONE: INTRODUCTION TO MULTIMEDIA SIGNALS

1.1 SIGNALS

A signal is a function that conveys information. Any quantity that exhibits variation in time and/or space and is therefore capable of conveying a message is a signal. Examples include audio, video, speech, image, communication and musical signals. There are two main types of signals: analog and digital.

An analog signal is a continuous signal whose amplitude can take any value within its range. In an analog signal, there is infinite number of possible values. If it is known that the output of a microphone that produces analog voltages in proportion to the intensity of sound ranges from 0 to 1mV, then the voltage at any instance of time can be 0.1002mV, 0.5432mV or in fact any of the infinite values between 0mV and 1mV.

A digital signal is a sequence of discrete values which can only take one out of a finite number of values in any instance of time. If the output of the microphone mentioned above is to be represented as a digital signal, then a decision must be made beforehand as to what the discrete values shall be such as 0.00mV, 0.01mV, 0.02mV in steps up to 1.00mV. This gives rise to 101 discrete values to which any instant value must be approximated.

1.2 COMPARISON BETWEEN ANALOG AND DIGITAL SIGNALS

i.Noise Rejection: The most important quality of a signal is its ability to convey information without losing the information to noise. Even when a signal is weak, it can easily be amplified if there is a means of rejecting the noise and amplifying only the signal. Sources of noise include transmission media, signal processing circuits and storage media. The most important advantage of a digital signal over an analog signal is the superior noise rejection capability of digital signals. A digital signal exists as a series of 1's and 0's that can easily be represented as light and darkness, black and white, 5 volts and 0 volt, or indeed, any contrasting physical characteristics. A noise character can easily be rejected as neither black nor white or neither 0 volt nor 5 volts.

In analog signals there are infinite possible values and so it is difficult to identify any value as noise and not a valid signal.

ii. Suitability for multimedia: All digital signals, whether they are carrying text information or video or audio or any other type of information, are always represented as a series of 1's and 0's. They can be transmitted the same way, filtered the same way, amplified or stored the same way. The media recognizes every bit as either 1 or 0. In analog signals the signals are all different in numerous ways. Therefore, Digital signals are suitable for multimedia while analog signals are not.

iii. Ease of generating the signal: Analog signals are much easier to generate than digital signals. A signal conveys information about a physical quantity which is naturally analog. For example, in an analog telephone system in a local loop, the sound is converted to electrical voltage by the microphone. The connecting wire sends it to the receiver where the voltage is converted back to sound by the speaker. There is no amplification, no filtering, no digitizing or any kind of signal processing. But in a digital telephone system of the same local loop, the voltage generated by the microphone must first of all be filtered, digitized and modulated before transmission. At the receiver, the signal must be demodulated, converted back from digital to analog (the human ear cannot interpret 1's and 0's to mean anything), and filtered before getting to the speaker, we can say that analog signals are much simpler than digital signals.

iv. Bandwidth requirement: Digital signals require much higher bandwidth than analog signals. Every bit in a digital signal be it 1 or 0 requires a cycle of the carrier frequency to transmit or store. For this reason an analog telephone system can allocate as low as 8 kHz per subscriber but a digital phone system requires a much higher bandwidth. However, sophisticated digital compression techniques are now available for reducing the bandwidth requirement of digital signals but they still need higher bandwidth than analog signals,

1.3 MULTIMEDIA

Multimedia refers to an electronically delivered combination of media including video, still images, audio and text in a way that can be accessed interactively.

Multimedia may be broadly divided into linear and nonlinear. Linear multimedia

does not need navigational control to view. Examples include cinema presentation. Nonlinear multimedia are interactive. Video games and computer based training are examples.

Applications of multimedia are in advertising, arts, education, entertainment, engineering, medicine and business. These applications are made possible by the fact that although multimedia signals vary greatly in their nature and in the way they are generated, they can all be digitized and converted into binary digits that are processed essentially the same way.

CHAPTER TWO:

MULTIMEDIA SIGNAL REPRESENTATION AND PROCESSING

2.1 TEXT REPRESENTATION

The standard way of representing text as a multimedia signal is by using ASCII Code. ASCII stands for American Standard Code for Information Interchange. Other text representation schemes in use today are all extensions of the ASCII code. Such extensions are required when there is a need to accommodate non English alphabets such as Greek, Korean, Japanese and Chinese writings. French and German also have alphabets that are not included in ASCII code.

ASCII is a 7-bit code which means that a total of $2^7 = 128$ characters can be represented. The lowest bit (extreme right) is bit one while the highest bit is bit seven. There is no general formula for determining the code for each character but the following patterns apply:

- i. The first 32 characters from 0000000 to 0011111 (0-31) are control characters such as ESC (escape), CAN (cancel), ACK (acknowledge), BS (back space). They cannot be printed. In summary, any ASCII code that starts with 00 is not printable.
- ii. All remaining 96 characters including blank space are printable. A character is printable if either bit 6 or bit 7 or both are binary digit 1.
- iii. In all capital letters, bit 6 is 0 while in small letters, bit 6 is 1. The difference between letter d and letter D is only in bit 6 which is 0 for letter D and 1 for letter d. When the code for letter m is known, the code for M is obtained by changing bit 6 from 1 to 0. This is how word processing programs convert from lower case to upper case and vice versa.
- iv. Numerical characters, capital letters and small letters are all arranged in ascending order such that the code for letter E is one higher than the code for letter D; the code for letter e is one higher than the code for letter d; the code for number 7 is one higher the code for number 6. All numerical characters come before the alphabets. This pattern allows sorting and indexing programs to easily use ASDCII code for each

character to obtain an alphabetical ordering in ascending or descending order as may be required.

Hexadecimal Coding

ASCII codes are 7-bit codes. It is common in computer science to translate binary codes to their hexadecimal equivalent in order to shorten the length of the code word. Hexadecimal code is a base 16 coding system that uses capital letters A to F in addition to the 10 numerical digits from 0 to 9. A binary code is easily converted to a hexadecimal code by dividing the binary code into groups of 4 bits starting from the lowest bit. Each group of 4 bits is then converted to their hexadecimal equivalent. For example, ASCII code for letter N is 1001110 in binary. To convert it to hexadecimal, start from bit1, the last bit and count 4 bits, then count the remaining 3 bits. You get 100 | 1110. The first group is 4 and the last group is E. Therefore, the ASCII code for letter N is 1001110 in binary and 4E in hexadecimal.

As discussed earlier, the code for letter n is obtained from that of N by changing bit 6 from 0 to 1. The code in binary is 1101110. When converted to hexadecimal we obtain 6E. Hence, given the code for a capital letter in hexadecimal, you can obtain the code for the equivalent small letter by increasing the first hexadecimal digit by 2. The second digit is the same for capital and for small letter. The code for M in hexadecimal is obtained by decreasing that of N by 1 to get 4D. That of m is 6D. The table below shows all the characters with their hexadecimal representation. There is no need to memorize the code but the pattern should be committed to memory.

ASCII Hex Symbol			ASCII Hex Symbol			ASCII Hex Symbol			ASCII Hex Symbol		
0	0	NUL	16	10	DLE	32	20	(space)	48	30	0
1	1	SOH	17	11	DC1	33	21	!	49	31	1
2	2	STX	18	12	DC2	34	22	"	50	32	2
3	3	ETX	19	13	DC3	35	23	#	51	33	3
4	4	EOT	20	14	DC4	36	24	\$	52	34	4
5	5	ENQ	21	15	NAK	37	25	%	53	35	5
6	6	ACK	22	16	SYN	38	26	&	54	36	6
7	7	BEL	23	17	ETB	39	27	'	55	37	7
8	8	BS	24	18	CAN	40	28	(	56	38	8
9	9	TAB	25	19	EM	41	29	)	57	39	9
10	A	LF	26	1A	SUB	42	2A	*	58	3A	:
11	B	VT	27	1B	ESC	43	2B	+	59	3B	;
12	C	FF	28	1C	FS	44	2C	,	60	3C	<

13 D CR	29 1D GS	45 2D -	61 3D =
14 E SO	30 1E RS	46 2E .	62 3E >
15 F SI	31 1F US	47 2F /	63 3F ?
ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol
64 40 @	80 50 P	96 60 `	112 70 p
65 41 A	81 51 Q	97 61 a	113 71 q
66 42 B	82 52 R	98 62 b	114 72 r
67 43 C	83 53 S	99 63 c	115 73 s
68 44 D	84 54 T	100 64 d	116 74 t
69 45 E	85 55 U	101 65 e	117 75 u
70 46 F	86 56 V	102 66 f	118 76 v
71 47 G	87 57 W	103 67 g	119 77 w
72 48 H	88 58 X	104 68 h	120 78 x
73 49 I	89 59 Y	105 69 i	121 79 y
74 4A J	90 5A Z	106 6A j	122 7A z
75 4B K	91 5B [	107 6B k	123 7B {
76 4C L	92 5C \	108 6C l	124 7C
77 4D M	93 5D]	109 6D m	125 7D }
78 4E N	94 5E ^	110 6E n	126 7E ~
79 4F O	95 5F _	111 6F o	127 7F .

Apart from ASCII codes hexadecimal representation of binary digits are commonly applied in MAC addresses, system id, microprocessor instruction codes and memory addressing.

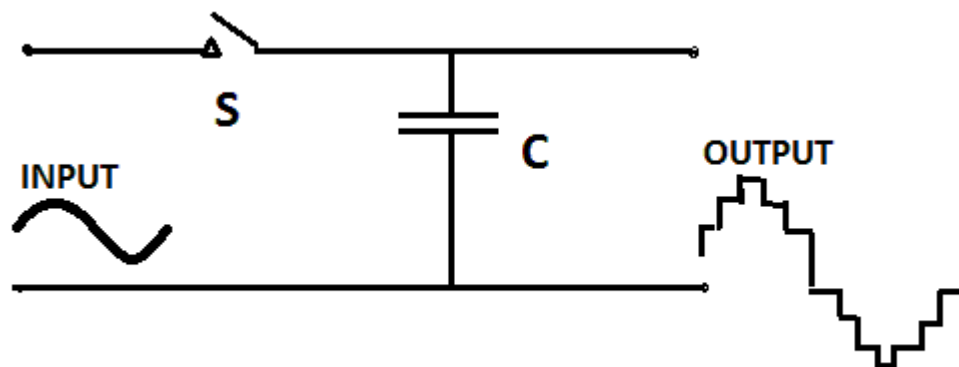
2.2 REPRESENTATION OF SOUND IN MULTIMEDIA

Sound is produced and transmitted by the vibration of materials. It is not possible to produce or transmit sound in a vacuum or free space. Sound is transmitted through the air or some other elastic medium as pressure waves that form around the vibrating material. These waves follow a pattern called waveform and occur repeatedly at regular intervals of time. The time interval is called period T . The number of periods per second is called frequency f . Mathematically, $f = 1/T$. T is measured in seconds while f is measured in hertz. The amplitude of the waveform is the measure of its intensity.

A sound wave is an analog signal. Using a mathematical technique called Fourier transform, any analog signal can be decomposed as a summation of single frequency sinusoidal signals. The range of frequencies which build up a given signal, i.e. the difference between the highest and the lowest frequency components is called the signal bandwidth.

To convert audio signal into a digital representation required for multimedia applications, the most common technique is pulse code modulation (PCM). The process involves setting a number of valid values in the amplitude axis of the waveform and measuring the amplitude of the wave a given number of times per second. The measurement at a given rate is referred to as sampling. The sampled values are rounded up to the closest valid value in the amplitude axis. The rounding up of samples is called quantization. The distance from one value to the next is called the quantization interval. Each quantization value is represented by a digital bit word. This way, an audio signal is represented digitally by a sequence of bit words which are the result of the sampling and quantization.

SAMPLE AND HOLD CIRCUIT



A **sample and hold circuit** basically consists of a capacitor C called hold capacitor connected to an input signal through a switch S . When the switch is turned on, the capacitor charges quickly to the input voltage. The switch is then turned off allowing the capacitor to retain the sampled voltage as a constant while the Analog-to-digital circuit converts this voltage to a bit word that represents that voltage level in binary. The period of time when the switch is open is called the hold time. It is also called the aperture time. The time the switch is closed is called the acquisition time. The two add up to the sample interval.

In practical sample-and-hold circuits, a buffer/amplifier is connected between the input and the switch. The buffer is designed to have low output impedance so that the acquisition time is very small. The hold capacitor is a high quality capacitor with low capacitance of about 1nF . The switch is usually the drain terminal of a Junction Field Effect Transistor (JFET) or Metal-Oxide Semiconductor Field Effect Transistor (MOSFET). The switch is controlled by the sample pulse generated by the sample clock. An output buffer is usually connected

between the hold capacitor and the output terminal in order to avoid loading the hold capacitor during the aperture time thereby maintaining a constant output at this time interval.

SAMPLING FREQUENCY

It has been noted earlier that every audio signal (and indeed any analog signal) can be decomposed into sinusoids of different frequencies. The number of digital samples per second executed in digitalizing the analog signal is called the sampling frequency. This frequency is determined by the **Nyquist rate or Nyquist frequency**. The Nyquist rate is twice the highest frequency in the analog signal. If a signal is sampled below the Nyquist rate the signal will be severely distorted. This distortion is called **Aliasing distortion**. Note that the higher the sampling frequency the larger the amount of digital data generated and this affects the cost of processing, transmission and storage. An audio signal can contain frequencies as high as 20KHz especially those from musical instruments. However, in telephone systems the audio signal must first be passed through a low pass filter to remove every frequency component above 4KHz before sampling. This low pass filter is called **Antialiasing filter**. The filter is connected at the input of the sample-and-hold circuit. The filtered signal has only frequency components below 4KHz. The Nyquist rate in this case is 8KHz. This is the sampling frequency used in telephone systems.

Problem

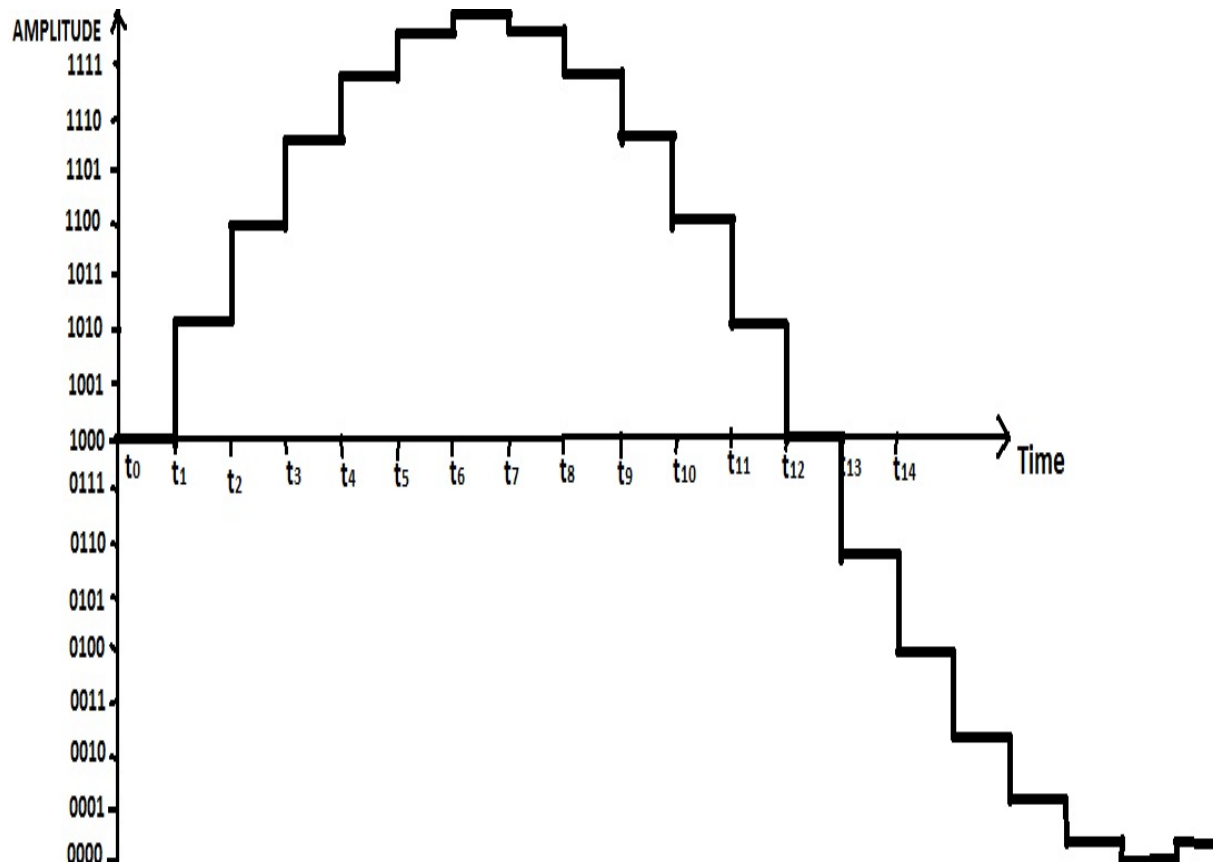
(1) The highest frequency component in Michael Jackson Thriller CD is found to be 20KHz. What is the Nyquist rate for this music for digital recording? If the lowest frequency in the music is 20Hz what is the bandwidth of the analog signal? For each of the frequencies listed below, state whether it is a suitable sampling frequency for the digital recording of this music and why in 2 – 3 sentences.

- (a) 20Hz
- (b) 20KHz
- (c) 30KHz
- (d) 44.1KHz
- (e) 45KHz
- (f) 200KHz
- (g) 400KHz

(2). In telephone systems your voice is sampled at 8KHz. What is the highest frequency component of your voice that can be heard by your caller? At what point in the digitizing process are the higher frequency components eliminated? Explain why 8KHz is used in telephone systems while music CD is recorded at 44.1KHz.

QUANTIZATION

We had earlier defined quantization as the process of rounding up the acquired samples of the analog signal to the closest valid values in the amplitude axis. This process is illustrated below.



Both the valid amplitude values 0001, 0010 etc. and the sampling intervals t_1 , t_2 etc. are determined beforehand. At t_1 , the acquired amplitude is 1010. That is recorded as the digital value of our signal at that time interval. At t_2 the value is 1100. At t_3 , the value is between 1101 and 1110 but is nearer to 1110 which is then recorded as the digital value at time t_3 . Similarly, all the other samples are rounded up to the nearest valid values on the amplitude axis. The more the number of valid values the more accurate our sampling will be. However, using more valid amplitude values increases the number of bits which must be processed, transmitted and stored in registers thereby increasing the cost. System designers must determine how many valid amplitude values they need in order to meet the required quality demanded by consumers. In telephone systems, 128 valid amplitude values are used resulting in 7 bits per sample.

DYNAMIC RANGE & QUANTIZATION NOISE

Dynamic range is defined as the ratio of the maximum input voltage to the smallest voltage level that can be quantized. It is also equal to the number of the valid amplitude values or number of quantization levels. In the example used to illustrate quantization above, the dynamic range is 16. In telephone systems it is 128. However, dynamic range is usually stated in decibel dB. Since the ratio is a voltage ratio, the dB value is given by the formula:

$$\text{Dynamic Range (in dB)} = 20 \times \log N$$

where N is the number of quantization intervals.

In our example, $N=16$.

When N is expressed in binary, the number of bits n is determined by the formula

$$N = 2^n$$

In a situation where n is not an integer, the next higher integer value is used. If N is 100, n is 7.

Usually $\log 2$ is approximated to 0.3 and since $\log 2^n = n \times \log 2$ the dynamic range is calculated as:

$$D.R = 20 \times n \times 0.3 \text{ dB}$$

In our example D.R is $20 \times 4 \times 0.3 = 24 \text{ dB}$

In digital telephone systems N is 128, dynamic range is $20 \times 7 \times 0.3 = 42 \text{ dB}$.

Max quantization noise = Quantization Interval / 2

Example

The signal from a cordless microphone ranges from 0 to 200mV. If the quantization interval 1mv, calculate:

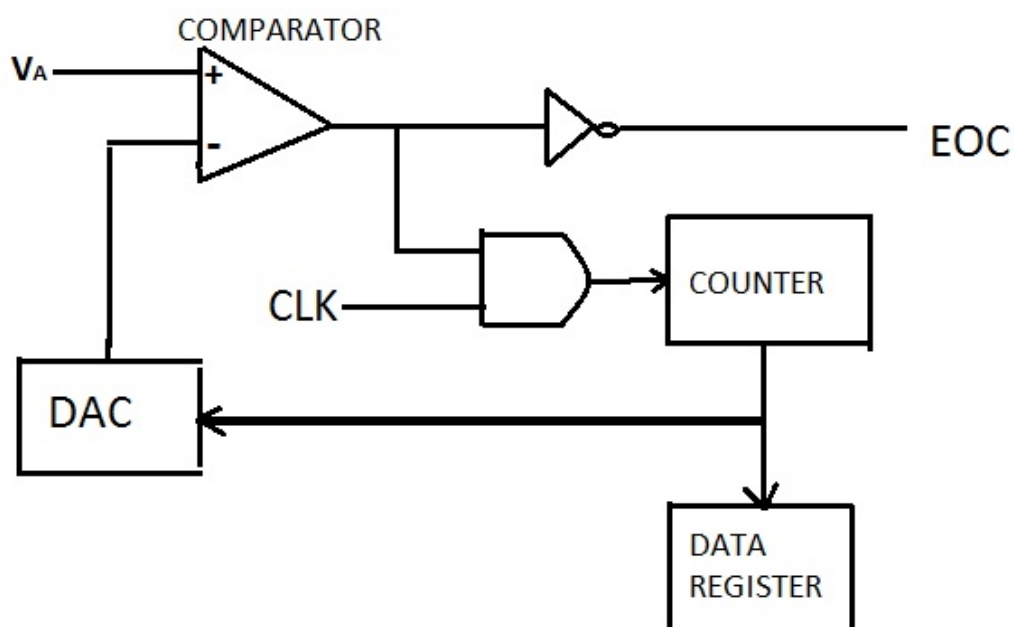
- i. The number of quantization levels
- ii. The number of bits required per sample
- iii. The dynamic range in decibels
- iv. The maximum quantization noise
- v. The sampling frequency if the sampling interval is $25\mu\text{s}$
- vi. Minimum bandwidth required to transmit the digital signal

Solution

- i. No of quantization levels = Input range / Quantization Interval
 $= 200\text{mV}/1\text{mV} = 200$
- ii. No of bits is n where $N = 2^n$; Now, 2^7 is 128, 2^8 is 256. Since n must be an integer, $n = 8$. Please note that if 200 were to be an integer power of 2 (like 256) then there would be no need to add anything to it.
- iii. D.R = $20 \times n \times 0.3\text{dB} = 48\text{dB}$
- iv. Max Quantization noise is $1\text{mV}/2 = 0.5\text{mV}$
- v. Sample Freq = $1/\text{Sampling interval} = 1/25\mu\text{s} = 10^6/25 = 40\text{KHz}$
- vi. Minimum Bandwidth = Sampling freq $\times$ No of bits per sample; $40\text{K} \times 8\text{bits} = 320\text{K bits/s}$

DIGITIZING

Analog-to-digital converter (ADC) is used to convert the result of the quantization process into a digital format (a series of 1's and 0's). This conversion process is known as digitizing.



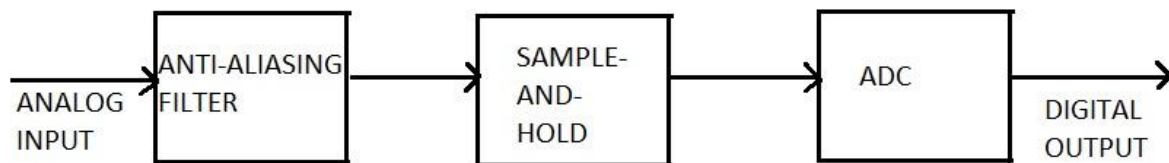
Ramp ADC

The major components of a ramp analog-to-digital converter are shown above. These components include the comparator, the counter, the data register and the digital-to-analog converter (DAC). The quantized analog signal is fed to the positive input of the comparator. The negative input of the comparator comes from the output of the DAC which is initially zero. The output of a comparator remains logic 1 for as long as the positive input is higher than the negative. Each clock pulse will be counted by the counter and will also increase the output of the DAC. The AND gate ensures that the counter stops when the DAC output equals the signal input. The output of the comparator now becomes logic 0. EOC (End Of Count) becomes logic 1 and causes the data register to record the last number from the counter which is the reset after a delay time necessary to register the count.

Other types of ADC include the successive approximation ADC and the dual-slope ADC. The successive approximation ADC is the most widely used.

The ADC circuit is often referred to as the encoder while the DAC is the decoder. These functions are usually combined in a single chip called codec (coder-decoder).

In summary the major components of a Pulse Code Modulation (PCM) system are shown below.



BANDWIDTH CONSIDERATIONS

The parameters that determine the bandwidth of a digital signal are the sampling rate and the number of bits per sample. These same parameters equally determine the quality of the digital signal. In telephone systems the sampling rate is 8KHz while the number of bits is 7. The minimum bandwidth is the product of the two figures. Therefore, a 56K modem is the minimum requirement that can handle voice data. Usually there are overheads for which reason 64Kbits/s or 64Kbs is allocated to voip applications in bandwidth resource allocation.

2.3 REPRESENTATION OF IMAGE AND VIDEO IN MULTIMEDIA

THE NATURE OF IMAGE AND VIDEO SIGNALS

An image signal is usually generated by a CCD camera. CCD stands for Charge Coupled Device. The CCD is a solid state chip consisting of thousands or millions of photosensitive cells arranged in a two-dimensional array. When light (photons) strike the CCD surface, the generated signal is an electronic analog representation of the image. The amplitude of the signal generated by each of the millions of cells is directly proportional to the intensity of the light falling on that cell just as the case of sound where the intensity of the sound determines the amplitude of the analog signal. However, in the case of video, the signal is generated by millions of cells rather than one microphone. If we imagine millions of cells arranged in a two-dimensional matrix across this page, then we can obtain a time-varying signal by recording serially the amplitude of the signal from each cell starting from the top of the page. We record the voltage of every cell on line 1, then line 2 and so on until we have scanned all the lines in the page. This is also the process followed in displaying a picture or a video on the screen. The first line is displayed completely from left to right, then the second line and third line and so on. In the case of a video there is motion involved so that the signal from each cell varies with time. All the lines must be scanned at least 2 times the rate of variation of the scene. In order to include colour information, the CCD camera must have output for each of the three primary colours of red, blue and green. For these reasons, a video signal has frequency components from zero to several million hertz whereas an audio signal ranges up to about 20,000 hertz. The consequences of this fact are as follows:

1. Graphic manipulation programs such as Photoshop and Corel Draw often require large memory and fast processors to function well.
2. Television channels require bandwidth of 6MHz whereas AM radio only needs 10KHz.
3. Digitalization of video signals requires rigorous compression processes which has led to various standard like MPEG-4 (MP4), H.263, VC-1, H.264
4. Pictures and graphics for email are usually saved in a compressed format such as JPEG.

AUDIO-VISUAL SIGNALS

The audio and video signals in multimedia are processed independently. They run in two different streams such that it is possible to notice the audio output out of synch with the video output after a video editing work. In television they are transmitted, received and

processed independently but a timing signal called the synch is transmitted along. The television sets receive the synch and use it to synchronize the video and the audio signals after they have been processed independently.

CHAPTER THREE: DIGITAL SATELLITE COMMUNICATIONS

3.1 SYNCHRONOUS ORBIT SATELLITES

Satellite communication is made possible by the placement of satellites in geostationary orbits (Geosynchronous or synchronous orbits). This means that the satellite is located at a fixed distance of about 35,700Km or 22,300 miles above the earth. By maintaining the same angular velocity as the earth, the satellite appears stationary to an observer on earth. At this attitude, the force of gravity provides the centripetal force that balances the centrifugal force generated by the motion of the satellite as it maintains a constant speed that ensures the same angular speed as the earth. This balance of forces keeps the satellite in orbit. On-board power thrusters are used periodically to reposition the satellite in order to correct for any drift from its orbit occasioned by gravitational attraction from other celestial objects other than the earth.

In satellite communication, the communication equipment can be found at 3 locations: the uplink (transmitter), the orbiting satellite and the downlink (receiver). The uplink and the downlink are located on the earth and are called earth stations or ground base stations. The orbiting satellite basically provides communication links between these earth stations. The earth station communicates with the orbiting satellite through a parabolic reflector and a receiver. On the satellite, the communication equipment are antennas and transponders. A transponder is an electronic system that performs reception, frequency translation and retransmission.

A geostationary satellite uses an equatorial orbit. This means that the point on the earth nearest to a geostationary satellite is on equator. For this reason the entire 360 degrees around the equator are allocated at the rate of 2 degrees per satellite. Such a satellite covers about a 3rd of the earth. It cannot cover areas around the North Pole and the South Pole.

A satellite can use a polar orbit moving from the North Pole to the South Pole and back again. Such a satellite can cover 100% of the earth surface but in this case the

satellite cannot be geostationary. In fact, every location on the earth is visible to the satellite twice a day. It is hence not suitable for communication. Its ability to observe events anywhere on earth makes it useful for military uses.

Apart from equatorial and polar orbits, inclined orbits are also used. Such satellites must be tracked by the receiving station. It is therefore not suitable for communication.

3.2 LOW EARTH ORBIT SATELLITES (LEO)

In order to accommodate more satellites, an orbit much lower than the geostationary orbit can be used. Low earth orbits range from as low as 250 miles above the earth. Satellites in LEO are not stationary with respect to an observer on earth. The satellite appears and disappears over the horizon. Systems that use such satellites deploy several satellites at a time. The subscriber is handed off from one satellite to the other to maintain constant communication. LEO satellites are arranged in such a way as to ensure that there is always a satellite visible to the subscriber at all times. An example of a LEO satellite system is the GPS (Global Positioning Satellite) which uses a constellation of 28 satellites orbiting the earth at 10,900 miles. At least 4 of these satellites are available to an observer on earth at any point in time. These satellites are used by GPS receivers to determine the exact location of any place on earth.

3.3 MERITS OF SATELLITE COMMUNICATION

1. It enables simultaneous communication over very large areas across the world.
2. Setting up earth stations is relatively easy. A large communication network can be set up in a matter of hours.

3. Mobile communication is easily achieved. An earth station such as MSAT can be mounted on a moving vehicle.
4. The cost does not increase much with more users or longer distances.
5. Difficult terrains like the sea and mountainous areas are easily covered.

3.4 DEMERITS OF SATELLITE COMMUNICATION

1. The communication path is very long. From earth station (sender) to satellite and from satellite to another earth station (receiver) can be up to 75,000Km.
2. The very long communication distance can introduce delays which is easily seen in television stations when two stations are communicating by satellite.
3. The cost can be very high for a single user as in the case of satellite phone set.
4. The very high technology involved in launching a satellite into orbit effectively limits the extent of participation that is available to low technology countries like Nigeria.

3.5: RECENT TECHNOLOGICAL ADVANCEMENT IN LEO SATELLITES

Why Low-Earth Orbit Satellites Are the New Space Race

By **Greg Ritchie | Bloomberg**

August 15, 2019 at 8:02 p.m. GMT+1

Elon Musk made headlines in 2018 when he launched his old car toward Mars aboard one of his Space Exploration Technologies Corp. rockets. He got less attention in May for his first step toward a potentially far more lucrative venture, when SpaceX launched the first 60 of a planned 12,000 satellites into low-Earth orbit (LEO). And Musk isn't the only one pouring money into the sector: Amazon.com Inc. Chief Executive Officer Jeff Bezos is one of several competitors with plans to send thousands of their own devices into the space just above our atmosphere. Nonetheless, doubts remain over whether these new satellite constellations will

provide returns on their substantial initial investments -- and even whether there's space in our sky for so many new devices.

1. What's Musk's plan?

In Musk's vision, SpaceX's Starlink satellites will form a network providing high-speed internet service to places where laying fiber-optic cables is not economically viable, such as emerging markets in Africa and Central Asia, along with remote spots in the U.S. They'd also provide specialized services for high-end business customers. Musk says Starlink, not the SpaceX cargo runs, are how the company will bankroll his ultimate goal of sending humans to Mars.

2. Why low Earth orbit?

LEO satellites operate at between 500 kilometers (310 miles) and 2,000 kilometers above the Earth's surface. That's far less than is typical for the 36,000 km (22,000 miles) height of so-called geostationary satellites, the traditional home of communications satellites. The main advantage of the lower orbit is lower latency.

3. What's latency?

Latency is the delay, usually measured in milliseconds, that occurs in a round-trip data transmission. Geostationary satellite systems have a median latency of nearly 600 milliseconds, making them an unsuitable replacement for cable or fiber systems. The lower orbits of LEO satellites, however, should result in latencies that are much closer to landline quality. OneWeb, a global communications company based in London that's in the process of creating a network of about 650 LEO satellites, recorded an average latency of 32 milliseconds in July. Speeds like that have made satellites of interest to financial trading firms and other companies that rely on having the fastest connections to trading exchanges.

4. Are LEO satellites new?

No. In fact, most of the Earth's approximately 2,000 active satellites are already in LEO. Iridium Communications Inc.'s network -- which allows voice and data communication from hand-held satellite phones through LEO satellites -- has been active since 1998 and currently has 66 satellites in operation. What's new is the sheer scale of recent proposals, with the big firms planning to launch satellites in the thousands. This ambition has been driven by technological developments in smaller satellites and reusable rockets, which have brought down costs and aroused investor interest.

5. Why do they need so many?

At higher altitudes, satellites can settle into a geostationary orbit, moving at a speed that matches the Earth's rotation and appearing to hover over a fixed spot. LEO satellites whiz around the planet at around 8 kilometers per second, completing a full circuit in between 90 and 120 minutes. That means they are only visible for a small part of their orbit to receivers on the ground. Thus, multiple satellites are necessary to establish a permanent internet connection, with one satellite passing duties to the next as it approaches the horizon. Musk has said sending as many as 800 satellites would ensure "moderate" coverage, though many more are needed for a truly global high-speed connection.

6. Who are the other big players?

While SpaceX has had plans for nearly 12,000 satellites approved by the Federal Communications Commission (FCC), so far the company has only done a single launch of 60 satellites back in May. Meanwhile, in early July Amazon asked for permission to launch 3,236 satellites through what it's calling Project Kuiper, creating a high-stakes battle between two of the world's most famous billionaires. OneWeb, meanwhile, hopes to start monthly rocket launches carrying 30 satellites

each by the end of the year to build an initial network of 648 units. It plans to provide full global commercial coverage by 2021 and partial service from as early as 2020. So far, it has launched six satellites. Other companies, such as traditional operators Inmarsat Plc and Eutelsat SA, are at a similarly early stage. And in China, the Hongyun Project is proposing a 156-satellite constellation by 2022.

7. What hurdles do these efforts face?

The issue of space debris is causing a lot of headaches. The so-called Kessler Effect -- named after NASA scientist Donald Kessler -- refers to the possibility that if LEO becomes too crowded, there will be collisions that create more debris, creating more collisions, until eventually huge tracts of space are no-go zones for spacecraft. SpaceX says 95% of its Starlink satellites will burn up in the Earth's atmosphere once they reach the end of their life cycle. In the mean time, the U.S. Air Force's 'Space Fence' -- a monitoring system designed to track satellites and debris -- is expected to go into operation in the fourth quarter of 2019.

8. Who regulates LEO?

In a sense, no one. But satellite operators have to get approval for their launch and orbit plans from their national communications regulator, and anyone planning to sell services to the U.S. needs to go before the FCC. The International Telecommunication Union coordinates the allocation of frequencies for communications.

9. Will the satellites be visible from Earth?

Potentially. Shortly after SpaceX launched its 60 satellites, astronomers said that some telescopes were picking up streaks of reflected sunlight, obscuring their view of the wider cosmos. Since around 13,000 low-Earth orbit satellites have been approved, a number that dwarfs the approximately 1,600 stars visible to the unaided human eye, astronomers voiced concern. In response, Musk has tweeted that "Starlink won't be seen by anyone unless looking very carefully" and that "we need to move telescopes [sic] to orbit anyway". He also said he had sent a note to his team about "albedo reduction," or cutting the proportion of light reflected from the spacecraft.

10. Are they going to make money?

Traditionally, satellites have been a big drain on capital and many projects never make any money. SpaceX has said completing their Starlink network may cost over \$10 billion, though Musk says it could bring in \$30 to \$50 billion per year once operational. Investors are right to be skeptical of promises of radically increased internet coverage and a transformed telecom market -- especially given the early stages of testing. Industry observers will be analyzing results from trial launches to gauge whether the big players are capable of delivering on their promises. Meanwhile, it's important to remember that despite all the fanfare LEO is far from the only show in town. Traditional operators like ViaSat Inc. and Eutelsat Communications SA are continuing to invest in more powerful geostationary satellites, collaborating with companies such as Facebook Inc. and Deutsche Telekom AG to beam broadband to rural areas and airplanes.

LEO and MEO broadband constellations mega source of consternation

by [Caleb Henry](#) — March 13, 2018

Divining what the stars hold in store for broadband megaconstellations

WASHINGTON — The world's biggest, best established satellite operators talk of broadband as an enormously lucrative opportunity.

But in truth, nothing is causing them more frustration.

Demand for ever-faster broadband internet connections is maxing out today's satellites, setting off an industry-wide stampede toward increasingly powerful high-throughput satellites (HTS).

While that might sound like a good thing, the rush to HTS is driving down bandwidth prices so fast that some fairly low-mileage satellites are struggling to keep up. A study by Northern Sky Research [found capacity prices dropped between 35 and 60 percent](#) over the last two years and will continue to decline into next year, with a rebound unlikely.

Seasoned operators, determined to stay ahead of the curve, are thinking twice before investing \$200 million or more in a geostationary broadband satellite designed to operate at least 15 years. Factoring in the two to four years it can take to build and launch a communications satellite, such a long-lived asset risks falling behind before it is in orbit.

Big, slow-moving geostationary satellite projects costing about as much as Paul Allen's superyacht or a Marvel superhero blockbuster have long dominated the satellite communications business. Fifteen-year satellites continue to do the trick when it comes to the biggest source of revenue for operators — transmitting television broadcast signals.

But broadband is a faster animal.

The dominance of big geostationary satellites is about to be tested by the emerging megaconstellations — nimble fleets of hundreds or even thousands of relatively small satellites orbiting the planet at significantly lower altitudes.

Cheaper to build, less costly to lose one or two, the low- and medium-Earth-orbit broadband satellites being built by OneWeb, SpaceX, Telesat and others promise nearly fiber-optic speeds and global coverage without the small-but-annoying lag that radio signals suffer during their 72,000-kilometer round trip to geostationary orbit and back. Though measured in milliseconds, latency is the bane of automated stock trades, hardcore gaming and Skype video chats.

Megaconstellations are not without their own sets of challenges, including uncertainty about how the fast-moving broadband market will evolve and questions about how cheaply satellite receivers can be put into the hands of the world's least served and close the digital divide (a popular *raison d'être* for many of these projects).

Planned for non-geostationary orbits, the megaconstellation satellites will orbit closer to Earth for faster connections. Those close orbits mean greater numbers are needed to cover the same surface area, driving the need for mass manufacturing and lots of launches. New types of constellations also means new regulatory approvals from nations where companies want to do business.

But these hurdles aren't stopping companies new and old from taking a swing at LEO and MEO broadband.

According to Northern Sky Research, at least 10 companies are planning to build broadband constellations of 100 satellites or more in non-geostationary-orbits. Most of these ventures anticipate having their first-generation constellations in orbit in the next five years.

Analysts agree that the advent of huge fleets of small satellites in non-geostationary orbits have the potential to change the paradigm for satellite internet. But no one knows exactly how since the megaconstellations aren't in orbit yet.

Some satellite operators are hedging their bets by taking stakes in the new constellations; others are buying large communications satellites equipped with features to make them more flexible and maximize throughput.

And some are doing both.

What's an operator to do?

Chris Quilty of Quilty Analytics says the usual metrics for predicting how fast a satellite will pay back the upfront investment "are out the window nowadays."

Armand Musey, president of the Summit Ridge Group telecom consultancy, agreed.

"The change is so rapid that five-, seven- or 10-year satellites for broadband are essentially all you'd want," Musey said "You don't want a 15- or 20-year broadband satellite any more than you want a 20-year-old laptop."

Avanti Communications provides a recent telling example. In December, the struggling British fleet operator said two of its still-young satellites — one launched within the past five years, the other within the past seven — can barely compete with only slightly younger systems.

WildBlue is another case in point. Boasting roughly 400,000 subscribers and \$200 million in revenue when Viasat bought the company for \$570 million in

2004, those numbers had shrunk to 100,000 subscribers and maybe \$80 million in revenue by 2014, according to Quilty. WildBlue was a “strategically important acquisition,” given ViaSat’s desire to be in the consumer broadband space, Quilty said, “but hard to see how they could have possibly generated a fair rate of return for the satellite.”

Some satellite operators are doubling down on geostationary broadband architectures.

Eutelsat is investing in a new satellite called Quantum that features beams that can change shape, size and power to better deliver capacity where and when it’s needed — a level of adaptability long sought in the satellite industry (Eutelsat ordered its first LEO smallsat last week, but the Tyvak-supplied nanosatellite is aimed at narrowband, Internet of Things applications, not broadband).

Global-IP, a recent startup led by a former Hughes Network Systems executive and advised by former AsiaSat CEO William Wade, ordered a 150-gigabits-per-second satellite from Boeing that’s launching in 2019 on a SpaceX Falcon 9 to connect users in Africa.

And dueling U.S. satellite broadband providers Hughes and ViaSat both have massive geostationary satellites launching in the early 2020s. — EchoStar 24/Jupiter-3 will deliver half a terabit per second of network capacity. ViaSat-3 expects to deliver at least twice that.

Other large operators, including Intelsat, SES and Telesat, are investing in constellations of non-geostationary-orbit satellite systems built for broadband connectivity.

The broadband bandwagon

OneWeb, founded in 2012 under the name WorldVu Satellites, took a big leap ahead in 2015 when it lined up \$500 million from such heavyweights as Airbus, Coca-Cola and Virgin Group.

SpaceX grabbed its share of 2015’s spotlight when news spread it was opening a Seattle factory to build some 4,500 broadband satellites for a constellation since dubbed Starlink.

In 2016, OneWeb raised another \$1.2 billion in an investment round led by SoftBank. SpaceX, focused on building and launching rockets, has refused to say much more about its constellation.

Meanwhile, OneWeb submitted its license application to U.S. regulators, touching off a spate of me-too filings from companies wanting the Federal Communications Commission to know that they also had plans for delivering broadband from non-geostationary orbits.

OneWeb's initial satellites are coming together at an Airbus factory in France before production of the balance of 900 satellites shifts to the purpose-built OneWeb Satellite factory soon to open in Florida.

SpaceX and Telesat already have prototype LEO broadband satellites in orbit. Telesat's Phase 1 LEO satellite launched Jan. 12 on India's PSLV rocket. SpaceX, on Feb. 22, orbited two experimental satellites, Tintin A and Tintin B, on a Falcon 9 rocket.

OneWeb's first satellites are slated to launch in May on an Arianespace-operated Soyuz rocket.

Come 2019, SpaceX and OneWeb expect to conduct regular launches to put their constellations in orbit.

Telesat plans to select a manufacturer for its constellation of 120 satellites later this year and start launching them in 2020.

LeoSat is working with Thales Alenia Space on a plan to produce a network of 78 to 108 satellites.

And more internet constellations are promised, such as China's 300-satellite Hongyan constellation, and Boeing's constellation of nearly 3,000 satellites.

Expectations of success

No one thinks every single proposed constellation is going to reach orbit, but most analysts agree one or more broadband constellation will ultimately pan out. "At least one, but less than five," ventured Quilty. "Is there room for two? I think there is."

People have different opinions on who the winners and losers will be.

Tim Farrar, president of the telecom analyst group TMF Associates, counts the medium-Earth-orbit O3b system now owned by SES as the first successful modern non-geostationary-orbit broadband system. O3b's first satellites launched in 2013 ahead of the current rush.

OneWeb, whose May launch will be of operational satellites instead of prototypes, is in second place, he said.

"At the moment there is a potential scramble for who is going to be the third player between Telesat, LeoSat and SpaceX," said Farrar. "It's probably going to be clear in the next 12 months which of those is out ahead."

The last four satellites in O3b's first-generation constellation of 20 satellites are scheduled to launch in 2019 on an Arianespace Soyuz. O3b mPower, a second-generation system of seven satellites bringing some 10 terabits of throughput, are slated to start launching in 2021.

OneWeb anticipates rolling out its global broadband service in 2019, starting with Alaska. SpaceX aims to offer limited service as soon as 2020 and Telesat is shooting for 2021.

Who cares who's first?

While OneWeb appears to be leading the pack, being first to market has its pros and cons, analysts said.

“Being the first to launch a new LEO broadband constellation has advantages, possibly to attract rural customers which don’t have any broadband connectivity or households with poor quality internet access,” said Khin Sandi Lynn, an industry analyst at market-foresight advisory firm ABI Research. “But it also has risk such as LTE coverage expansion or possibility of 5G networks in future.”

AT&T, T-Mobile and Sprint announced at the Mobile World Congress last month that they will start rolling out 5G, or fifth-generation wireless networks, to limited U.S. cities later this year. The long-awaited upgrade is expected to be 100 to 1,000 times faster than current 4G service.

Lynn noted that early this year the second-largest cable operator in the U.S., Charter Communication, announced it will be testing the delivery of 5G service to rural areas — one of the markets that LEO broadband constellations aim to serve.

Miscalculations about market size and equipment costs can be deadly, as ICO and Teledesic found out during the ill-fated satellite broadband gold rush of the 1990s.

In the race to equip corporate road warriors with go-anywhere satellite phones, Iridium beat rival Globalstar to orbit by about a year. Iridium also beat Globalstar to bankruptcy court, filing for Chapter 11 in 1999. Globalstar hung on until 2002.

“I don’t think it’s important to be first. It’s important to get it right,” said Musey.

Both Iridium and Globalstar eventually emerged from bankruptcy. Globalstar finished deploying its second-generation constellation in 2013. Iridium, which got off to a later start the second time around, is well on its way to having all of its new satellites in orbit by year’s end.

Teledesic and ICO weren’t so lucky. Both collapsed short of the finish line with no such phoenix-style resurrection.

“This whole idea of a first mover advantage is a bit overstated,” Musey said. “It works in certain consumer brand applications, like opening a McDonald’s in a neighborhood, but I don’t think that’s ever been an issue in the satellite industry.”

Rather than being an advantage, first movers could find competitors drafting behind them, taking advantage of their trailblazing in technology and regulatory development to speed their own systems, Quilty said.

Justifying megaconstellations

The greater question as to who will succeed lies in how much demand their constellations will actually generate.

“If you make a comparison with the 1990s, the biggest issue was that the market forecasts were dramatically wrong,” said Farrar. “It wasn’t that they couldn’t build the satellites or handsets. They made the technology work, but they didn’t find the market that they expected. That’s always the challenge for something big and new, and it doesn’t matter whether you are building LEOs or GEOs. When you are trying to get into a new market, it’s very difficult to come up with an accurate market forecast.”

OneWeb and SpaceX have both highlighted consumer broadband with an emphasis on connecting large, unreached populations to the internet as a central purpose of their constellations. Those populations will need low-cost, easy-to-install user terminals, otherwise prospective customers will find themselves priced out.

Iridium and Globalstar bet big on a large satellite-telephony market — a much larger market than actually materialized, thanks in part to a faster-than-expected expansion of cellular networks to previously underserved regions.

Sima Fishman, managing director of Euroconsult USA, said the current generation of LEO constellations face similar issues as some of their 1990s forebears, whose ambitious plans sparked a reusable launch vehicle boom that went bust (None of these ventures ever reached space, but many of them managed to raise millions and produce at some hardware all on the strength of good slide decks and non-binding letters of intent from Teledesic and the like).

“The most significant hurdle from our perspective is establishing credibility that demand will exist at the prices necessary to make the business case work,” she said.

A good ground game

Perhaps the biggest variable in calculating demand is figuring out what consumers will have to pay for the user terminal—the receiver and antennas customers will use to connect to the constellation.

In contrast to geostationary satellites that always appear to hover at the same point in the sky, LEO broadband constellations will require antennas capable of tracking multiple satellites as they orbit overhead. Such antennas exist. In use for more than a decade, phased-array antennas commonly have no moving parts, relying instead on electronically steered beams to communicate with satellites. The technology is proven, but the cost is sky high.

For cheap, abundant satellite broadband to succeed, consumers will need cheap but capable antennas. Companies such as Kymeta and Phasor Solutions say more affordable flat-panel antennas are coming soon, but analysts are quick to note

that they are not here yet — at least not at price points acceptable to the average consumer.

“With the enormous amount of capacity coming online, the only plausible way that it can be absorbed is that there is a large consumer adoption,” said Musey. “The only way you can have a large consumer adoption is if you have reasonably priced ground equipment. That’s particularly true in developing countries, but it’s also true in the U.S. and most advanced countries. Consumers are not going to pay \$50,000 or \$100,000 or more for a tracking antenna. To hit a consumer price point you need to have flat panel antennas at very low price points.”

In 2015, when Elon Musk first unveiled SpaceX’s broadband constellation plan, he said the flat-panel antennas customers will need to link up with the system would be inexpensive enough — \$100 to \$300 —to be affordable the world over. Last month, SpaceX acknowledged to *The Wall Street Journal* that it doesn’t yet know how much the terminals will cost.

OneWeb has said it is working with chipmaker Qualcomm, aOneWeb investor, on low-cost antennas.

Telesat, in an interview with *SpaceNews* last October, revealed only that the success of the Telesat LEO constellation “is not dependent on a big leap in ground terminal performance and capabilities.”

Consumer confidence

Farrar is among analysts who doubt consumer broadband will be the biggest application for LEO constellations. Bridging the digital divide is a laudable goal, but backhaul — using satellites to help cellular networks to increase coverage and improve service — is an early market they can dominate with or without cheap antennas.

“If you are backhauling from a cell tower, then if a terminal costs a few thousand dollars, that’s not a fundamental barrier to deploying it because you may be spending \$1,000 a month on capacity,” Farrar said. “If you are providing home broadband to someone for \$70 a month, then a terminal that costs a few thousand dollars makes it completely untenable.”

Not every operator is counting on consumer broadband for success. Intelsat, which tried unsuccessfully to merge with OneWeb last year, is a reseller of OneWeb capacity for almost every market but consumer broadband — in-flight connectivity, maritime, government networks, driverless cars and more. LeoSat, likewise, is targeting corporations instead of consumers.

Changing tides

Regardless of whether the non-geostationary-orbit (NGSO) constellations succeed, they are already having an impact on the satellite industry.

From 2014 to 2017, geostationary satellite orders were well below average, dropping into the teens before falling last year to just seven orders worldwide as operators hedge their bets on what technology is best.

In addition to the investments Intelsat, SES and Telesat are making in NGSO ventures, Asia's largest geostationary satellite operator, Sky Perfect JSAT took a stake in LeoSat last year. ViaSat, meanwhile, has filed FCC paperwork for a constellation of up to 24 satellites operating in medium Earth orbit.

"A lot of satellite operators are flummoxed by the correct choice, and they are taking steps to stay engaged in NGSO activity," said Quilty. "There's only a handful of players that seem to have adopted a wholesale wait and see approach."

"[I]ndustry players seem to be recognizing that the megaconstellation approach to capacity expansion represents a sea-change in the economics of the satellite industry," said Fishman. "We see this recognition across the traditional value chain, from manufacturing and launch through operators and service providers, as well as customers. We also see stakeholders such as insurance providers, regulatory agencies/spectrum administrators, and the financial community sorting out the implications."

Manufacturers are angling for constellation construction contracts by promoting new smallsat platforms. Launch providers are designing adapters and deployers for constellations, or building new rockets specifically sized for dedicated smallsat missions. And operators of ground-based satellite gateways are installing new antennas around the world to provide turn-key solutions for constellation operators.

Yet even as the entire satellite industry pivots to LEO broadband as the next big thing, no one knows for sure what the stars have in store for today's constellations.

"It's not an easy business," Farrar said. "All of this is complicated."

This article originally appeared in the March 12, 2018 issue of SpaceNews magazine.

3.6 MATHEMATICAL DETAILS IN SATELLITE MOTION

How to Calculate a Satellite's Speed around the Earth

In space, gravity supplies the centripetal force that causes satellites (like the moon) to orbit larger bodies (like the Earth). Thanks to physics, if you know

the mass and altitude of a satellite in orbit around the Earth, you can calculate how quickly it needs to travel to maintain that orbit.

A particular satellite can have only one speed when in orbit around a particular body at a given distance because the force of gravity doesn't change. So what's that speed? You can calculate it with the equations for centripetal force and gravitational force. For a satellite of a particular mass, m_1 , to orbit, you need a corresponding centripetal force:

$$F_C = \frac{m_1 v^2}{r}$$

This centripetal force has to come from the force of gravity, so

$$\frac{Gm_1m_2}{r^2} = \frac{m_1 v^2}{r}$$

You can rearrange this equation to get the speed:

$$v = \sqrt{\frac{Gm_2}{r}}$$

This equation represents the speed that a satellite at a given radius must travel in order to orbit if the orbit is due to gravity. The speed can't vary as long as the satellite has a constant orbital radius — that is, as long as it's going around in circles. This equation holds for any orbiting object where the attraction is the force of gravity, whether it's a human-made satellite orbiting the Earth or the Earth orbiting the sun. If you want to find the speed for satellites that orbit the Earth, for example, you use the mass of the Earth in the equation:

$$v = \sqrt{\frac{Gm_E}{r}}$$

Here are a few details you should note on reviewing the orbiting speed equation:

- **You have to use the distance from the center of the Earth, not the distance above Earth's surface, as the radius.** Therefore, the distance you use in the equation is the distance between the two

orbiting bodies. In this case, you add the distance from the center of the Earth to the surface of the Earth, 6.38×10^6 meters, to the satellite's height above the Earth.

- **The equation assumes that the satellite is high enough off the ground that it orbits out of the atmosphere.** That assumption isn't really true for artificial satellites; even at 400 miles above the surface of the Earth, satellites do feel air friction. Gradually, the drag of friction brings them lower and lower, and when they hit the atmosphere, they burn up on re-entry. When a satellite is less than 100 miles above the surface, its orbit decays appreciably each time it circles the Earth. (Look out below!)
- **The equation is independent of mass.** If the moon rather than the artificial satellite orbited at 400 miles and you could ignore air friction and collisions with the Earth, it would have to go at the same speed as the satellite in order to preserve its close orbit (which would make for some pretty spectacular moonrises).

Human-made satellites typically orbit at heights of 400 miles from the surface of the Earth (about 640 kilometers, or 6.4×10^5 meters). What's the speed of such a satellite? All you have to do is put in the numbers:

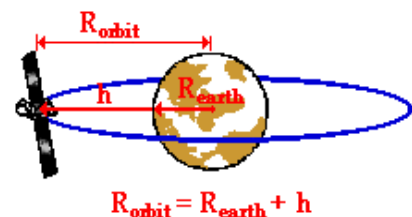
$$v = \sqrt{\frac{Gm_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2)(5.98 \times 10^{24} \text{ kg})}{(6.38 \times 10^6 \text{ m}) + (6.40 \times 10^5 \text{ m})}} \approx 7.54 \times 10^3 \text{ m/s}$$

This converts to about 16,800 miles per hour.

A geosynchronous satellite is a satellite that orbits the earth with an orbital period of 24 hours, thus matching the period of the earth's rotational motion. A special class of geosynchronous satellites is a geostationary satellite. A geostationary satellite orbits the earth in 24 hours along an orbital path that is parallel to an imaginary plane drawn through the Earth's equator. Such a satellite appears permanently fixed above the same location on the Earth. If a geostationary satellite wishes to orbit the earth in 24 hours (86400 s), then how

high above the earth's surface must it be located? (Given: $M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$,
 $R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$)

Just as in the previous problem, the solution begins by the identification of the known and unknown values. This is shown below.



Given/Known:

$$T = 86400 \text{ s}$$

$$M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$$

$$R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$$

$$G = 6.673 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

Unknown:

$$h = ???$$

The unknown in this problem is the height (**h**) of the satellite above the surface of the earth. Yet there is no equation with the variable **h**. The solution then involves first finding the radius of orbit and using this R value and the R of the earth to find the height of the satellite above the earth. As shown in the diagram at the right, the radius of orbit for a satellite is equal to the sum of the earth's radius and the height above the earth. The radius of orbit can be found using the following equation:

$$\frac{T^2}{R^3} = \frac{4 \cdot \pi^2}{G \cdot M_{\text{central}}}$$

The equation can be rearranged to the following form

$$R^3 = [(T^2 \cdot G \cdot M_{\text{central}}) / (4 \cdot \pi^2)]$$

The substitution and solution are as follows:

$$R^3 = [((86400 \text{ s})^2 \cdot (6.673 \times 10^{-11} \text{ N m}^2/\text{kg}^2) \cdot (5.98 \times 10^{24} \text{ kg})) / (4 \cdot (3.1415)^2)]$$

$$R^3 = 7.54 \times 10^{22} \text{ m}^3$$

By taking the cube root of $7.54 \times 10^{22} \text{ m}^3$, the radius can be determined to be

$$R = 4.23 \times 10^7 \text{ m}$$

The radius of orbit indicates the distance that the satellite is from the center of the earth. Now that the radius of orbit has been found, the height above the earth can be calculated. Since the earth's surface is 6.37×10^6 m from its center (that's the radius of the earth), the satellite must be a height of

$$4.23 \times 10^7 \text{ m} - 6.37 \times 10^6 \text{ m} = \mathbf{3.59 \times 10^7 \text{ m}}$$

above the surface of the earth. So the height of the satellite is $\mathbf{3.59 \times 10^7 \text{ m}}$.

Investigate!

There are hundreds of artificial satellites orbiting the Earth. A list of geostationary satellites can be found at <http://www.satsig.net/sslist.htm>. Use the **Satellite Information** widget below to explore the various properties - speed, height, orbital path, etc. - of any existing satellite. Simply type in the name (correctly) of the satellite and click on the **Get Information** button.

List of Satellites in Geostationary Orbit

If you access this "list of satellites in orbit" page, please help by sending updates and changes to me: [Eric Johnston](#)

List below updated 26 April 2020

Total number of active satellites =

534 (26 April 2020) This includes some satellites with an indicated high inclination angle.

517 (12 Dec 2019) This includes some satellites with an indicated high inclination angle.

448 (17 December 2018)

446 (3 Oct 2018)

450 (8 April 2018)

449 (10 Jan 2018)

447 (10 Aug 2017)

443 (11 Feb 2017)

419 (20 November 2015)

427(9 July 2016)

429(16 Aug2016)

Name refers to the official spacecraft designation, 5 digits and one or two letters. Made up as follows: 2 digits=year of launch so 12=2012; next 3 digits= day of year so 061 is 2nd of March, followed by letter(s) so A= is first item off the rocket, B= second item off the rocket, etc. Z= 26th, AB is 27th item off the rocket.

Common name is what companies use to give it an easy (?) to remember name. They are changed from time to time, for example if the satellite leased to a new customer or marketing people think of some new idea.

Geo Orbit position is the longitude position around the geostationary orbit. The satellites are all approximately fixed in the sky above the equator. **Negative orbit position numbers are degrees West** from Greenwich meridian, like Spain, Portugal, Atlantic, West West Africa, Canada, USA, Central and South America. **Positive numbers are degrees East**, like Central Europe, East Africa, Middle East, Asia, China, Japan and Australia.

Inclination: Some older satellite, whose north-south station-keeping fuel has run out are in inclined orbits. The inclination angle is shown. This means they move north-south across the equatorial plane on a daily basis, by plus or minus the amount shown. Tracking earth station antennas are used to follow these satellites. Example of tracking antenna methods and mechanisms: <https://www.s3sat.com/index.php/satsio>

Name	Common name	Orbit	Inc
11056A	INTELSAT 18 (IS-18)	-180.0	0.0
00001A	USA 148	-179.8	8.3
12061B	YAMAL 300K	-177.0	0.0
13036A	MUOS-2	-177.0	2.7
09008A	NSS-9	-176.9	0.0
13004A	TDRS 11	-174.4	3.9
02055A	TDRS 10	-171.2	6.8
91054B	TDRS 5	-167.5	14.3
00024A	USA 149 (DSP 20)	-165.5	11.5
02057A	NSS-6	-164.8	0.0
17004A	SBIRS GEO-4 (USA 273)	-159.4	3.9
18079A	AEHF-4 (USA 288)	-150.1	4.0
02044A	HISPASAT 30W-4	-143.0	1.0
14043B	USA 254	-141.1	0.4
00081B	AMC-8 (GE-8)	-139.0	0.5
06054B	AMC-18	-139.0	0.0
05046B	SPACEWAY 2	-138.9	0.1
18022A	GOES 17	-137.2	0.0
97046A	INTELSAT 5 (IS-5)	-137.0	5.6
13041A	WGS F6 (USA 244)	-135.2	0.0
99060A	AMC-4 (GE-4)	-134.9	0.0
02029A	EXPRESS-A4	-134.0	7.8
05041A	GALAXY 15 (G-15)	-133.0	0.0
01011A	EUTELSAT 133 WEST A	-132.8	1.8
04017A	AMC-11 (GE-11)	-131.0	0.1

96054A	AMC-1 (GE-1)	-130.9	3.8
03008A	USA 167	-130.0	6.5
17026A	SES-15	-129.1	0.0
03013B	GALAXY 12 (G-12)	-129.0	1.4
08063A	CIEL-2	-128.8	0.0
10008A	GOES 15	-128.1	0.2
03044A	GALAXY 13 (HORIZONS 1)	-127.0	0.0
05030A	GALAXY 14 (G-14)	-125.0	0.1
08038B	AMC-21	-124.9	0.0
08024A	GALAXY 18 (G-18)	-123.0	0.0
03034A	GALAXY 23 (G-23)	-121.0	0.0
03012A	USA 169 (MILSTAR-2 4)	-120.0	9.0
04016A	DIRECTV 7S	-119.1	0.0
10010A	EHOSTAR 14	-118.9	0.0
02006A	EHOSTAR 7	-118.8	0.0
07009A	ANIK F3	-118.7	0.0
10065A	HYLAS 1	-118.3	0.0
16038B	EUTELSAT 117 WEST B	-117.0	0.0
13012A	EUTELSAT 117 WEST A	-116.8	0.0
13058A	SIRIUS FM-6	-116.1	0.0
06049A	XM-4 (BLUES)	-115.3	0.0
11059A	VIASAT-1	-115.1	0.0
15010B	EUTELSAT 115 WEST B	-114.9	0.0
12075B	MEXSAT 3	-114.8	0.0
15056A	MORELOS 3	-113.1	4.9
06020A	EUTELSAT 113 WEST A	-113.0	0.0
97065A	USA 134	-112.2	10.2
06054A	WILDBLUE-1	-111.2	0.0
04027A	ANIK F2	-111.1	0.0
09035A	TERRESTAR-1	-111.0	2.5
06003A	EHOSTAR 10	-110.2	0.0
02023A	DIRECTV 5 (TEMPO 1)	-110.1	0.0
08035A	EHOSTAR 11	-110.0	0.0
99059A	TELSTAR 12 (ORION 2)	-109.2	3.2
96022A	MSAT M1	-107.5	9.2
13014A	ANIK G1	-107.3	0.0
05036A	ANIK F1R	-107.3	0.0
00076A	ANIK F1	-107.3	0.0
12035A	EHOSTAR 17	-107.1	0.0
95019A	AMSC 1	-106.5	11.2
04041A	AMC-15	-105.0	0.1
16041A	MUOS-5	-105.0	7.4
17063A	SES-11 (EHOSTAR 105)	-104.9	0.0
09033A	GOES 14	-104.1	0.2
96053A	INMARSAT 3-F2	-103.4	4.8
11035A	SES-3	-103.0	0.0
15026A	DIRECTV 15	-102.8	0.0
09075A	DIRECTV 12	-102.8	0.0
07032A	DIRECTV 10	-102.7	0.0
10005A	SDO	-101.7	30.2
10061A	SKYTERRA 1	-101.3	2.2
06043A	DIRECTV 9S	-101.2	0.0
10016A	SES-1	-101.0	0.0
19034A	AT&T T-16	-100.9	0.0
05019A	DIRECTV 8	-100.7	0.0
12009A	MUOS-1	-100.0	2.5
14078B	DIRECTV 14	-99.2	0.0
08013A	DIRECTV 11	-99.2	0.0
06023A	GALAXY 16 (G-16)	-99.0	0.0
08039A	INMARSAT 4-F3	-98.0	3.1
16079A	EHOSTAR 19	-97.1	0.0
08045A	GALAXY 19 (G-19)	-97.0	0.0
94084A	USA 107 (DSP 17)	-96.5	14.5
03020A	HELLAS-SAT 2	-95.2	1.8

02030A	GALAXY 3C (G-3C)	-95.0	0.0
16035A	INTELSAT 31 (IS-31)	-95.0	0.0
14062A	INTELSAT 30 (IS-30)	-95.0	0.0
07036A	SPACEWAY 3	-94.9	0.0
19051A	AEHF-5 (USA 292)	-94.2	6.1
99071A	GALAXY 11 (G-11)	-93.1	1.1
08016A	ICO G1	-92.8	3.6
00046A	BRASILSAT B4	-92.0	3.9
12026A	NIMIQ 6	-91.1	0.0
07016B	GALAXY 17 (G-17)	-91.0	0.0
95060A	USA 115 (MILSTAR-1 2)	-90.0	13.2
05022A	GALAXY 28 (G-28)	-89.0	0.0
13075A	TKSAT-1 (TUPAC KATARI)	-87.2	0.0
11049A	SES-2	-87.0	0.0
02015B	ASTRA 3A	-86.8	5.8
99027A	NIMIQ 1	-86.5	1.0
09034A	SIRIUS FM-5	-86.1	0.0
10053A	XM-5	-85.2	0.0
05008A	XM-3 (RHYTHM)	-85.1	0.0
04048A	AMC-16	-85.0	0.1
97002A	AMC-2 (GE-2)	-84.8	6.3
16082B	STAR ONE D1	-84.0	0.0
00067A	AMC-6 (GE-6)	-83.0	0.1
08044A	NIMIQ 4	-82.0	0.0
15054B	ARSAT 2	-81.0	0.0
15026B	SKY MEXICO-1	-78.8	0.0
11054A	QUETZSAT 1	-77.0	0.1
10006A	INTELSAT 16 (IS-16)	-76.2	0.0
16071A	GOES 16	-75.2	0.0
12062A	STAR ONE C3	-75.0	0.0
17023B	SGDC	-74.8	0.1
14011A	AMAZONAS 4A	-73.9	0.1
94070A	ASTRA 1D	-73.0	9.0
09050A	NIMIQ 5	-72.7	0.0
97050A	AMC-3 (GE-3)	-72.0	2.7
14062B	ARSAT 1	-71.8	0.0
15034B	STAR ONE C4	-70.0	0.1
08018B	STAR ONE C2	-70.0	0.1
17029A	VIASAT-2	-69.9	0.0
17014A	EHOSTAR 23	-67.9	0.0
12019A	AEHF-2 (USA 235)	-67.5	1.9
17017A	SES-10	-66.9	0.0
16014A	EUTELSAT 65 WEST A	-65.2	0.1
07056A	STAR ONE C1	-65.0	0.0
11021A	TELSTAR 14R	-63.0	0.0
18059A	TELSTAR 19V	-63.0	0.0
02011A	TDRS 9	-62.4	7.4
10034A	EHOSTAR 15	-61.6	0.0
12065A	EHOSTAR 16	-61.5	0.0
16039B	EHOSTAR 18	-61.4	0.0
13006A	AMAZONAS 3	-61.0	0.0
17053A	AMAZONAS 5	-61.0	0.0
09054A	AMAZONAS 2	-61.0	0.1
12045A	INTELSAT 21 (IS-21)	-58.0	0.0
15039A	INTELSAT 34 (IS-34)	-55.5	0.0
15005A	INMARSAT 5-F2	-55.0	0.0
98006B	INMARSAT 3-F5	-54.0	4.7
12057A	INTELSAT 23 (IS-23)	-53.0	0.0
13024A	WGS F5 (USA 243)	-52.5	0.0
01039A	INTELSAT 902 (IS-902)	-50.4	0.5
00043A	INTELSAT 9 (IS-9)	-50.0	5.6
88091B	TDRS 3	-49.1	14.2
01033A	USA 159 (DSP 21)	-49.0	10.7
18012B	SES-14	-47.5	0.0

93003B	TDRS 6	-46.1	14.0
02016A	INTELSAT 903 (IS-903)	-45.2	2.1
09064A	INTELSAT 14 (IS-14)	-45.0	0.0
17007B	INTELSAT 32E (IS-32E)	-43.2	0.0
07044B	INTELSAT 11 (IS-11)	-43.0	0.0
03040A	USA 170	-42.4	5.8
14004A	TDRS 12	-41.2	4.7
13026A	SES-6	-40.5	0.0
94009A	USA 99 (MILSTAR-1 1)	-38.8	14.1
09009A	TELSTAR 11N	-37.6	0.0
05003A	NSS-10 (AMC-12)	-37.4	0.9
17006A	HISPASAT 36W-1	-36.0	0.0
17041A	INTELSAT 35E (IS-35E)	-34.5	0.0
18033B	HYLAS 4	-33.5	0.1
08034A	INTELSAT 25 (IS-25)	-31.5	0.0
18023A	HISPASAT 30W-6	-30.0	0.1
10070A	HISPASAT 30W-5	-30.0	0.1
06007A	SPAINSAT	-30.0	0.0
02007A	INTELSAT 904 (IS-904)	-29.5	1.3
19080B	INMARSAT GX5	-27.5	0.1
03007A	INTELSAT 907 (IS-907)	-27.5	0.1
01024A	INTELSAT 901 (IS-901)	-27.5	0.1
17078A	ALCOMSAT 1	-24.8	0.1
02027A	INTELSAT 905 (IS-905)	-24.5	1.2
99063A	UFO 10 (USA 146)	-22.9	7.8
12007A	SES-4	-22.0	0.0
18012A	AL YAH 3	-20.1	0.0
02019A	NSS-7	-20.0	4.1
17059A	INTELSAT 37E (IS-37E)	-18.0	0.0
08030A	SKYNET 5C	-17.8	0.1
12061A	LUCH 5B	-16.3	5.6
15002A	MUOS-3	-15.5	2.9
15068A	TELSTAR 12V	-15.0	0.0
15048A	EXPRESS-AM8	-14.0	0.0
11048A	COSMOS 2473	-13.5	0.1
01042A	EUTELSAT 12 WEST B	-12.5	0.9
09068A	WGS F3 (USA 211)	-12.0	0.0
17047A	TDRS 13	-11.4	5.7
09007A	EXPRESS-AM44	-11.0	0.0
15039B	EUTELSAT 8 WEST B	-8.0	0.1
11051A	EUTELSAT 7 WEST A	-7.3	0.1
10037A	NILESAT 201	-7.0	0.1
06033B	SYRACUSE 3B	-5.2	0.0
19067A	EUTELSAT 5 WEST B	-5.0	0.1
02035A	EUTELSAT 5 WEST A	-5.0	0.4
08022A	AMOS-3	-4.1	0.1
14046A	ASIASAT 8	-3.9	0.0
15010A	ABS-3A	-3.0	0.0
99009B	SKYNET 4E	-1.2	10.8
04022A	INTELSAT 10-02	-1.0	0.0
09058B	THOR 6	-0.8	0.0
08006A	THOR 5	-0.7	0.0
15022A	THOR 7	-0.7	0.0
15034A	METEOSAT-11 (MSG-4)	-0.3	0.5
17038A	BULGARIASAT-1	1.9	0.0
10037B	RASCOM-QAF 1R	2.9	0.0
14030A	EUTELSAT 3B	3.1	0.1
05049B	METEOSAT-9 (MSG-2)	3.5	4.3
10039A	AEHF-1 (USA 214)	4.1	2.6
07057A	ASTRA 4A (SIRIUS 4)	4.8	0.0
12036A	SES-5	5.0	0.1
07046A	WGS F1 (USA 195)	6.0	0.0
13022A	EUTELSAT 7B	7.0	0.1
19034B	EUTELSAT 7C	7.0	0.0

97008A	USA 130 (DSP 18)	8.4	13.5
10069A	EUTELSAT KA-SAT 9A	9.0	0.0
16005A	EUTELSAT 9B	9.0	0.1
12035B	METEOSAT-10 (MSG-3)	9.2	1.0
09016A	EUTELSAT 10A	10.0	0.1
17032A	ECHOSTAR 21	10.3	5.6
19048A	COSMOS 2539	12.0	0.0
08065A	EUTELSAT HOT BIRD 13C	13.0	0.1
06032A	EUTELSAT HOT BIRD 13B	13.0	0.1
06007B	EUTELSAT HOT BIRD 13E	13.0	0.1
10021B	COMSATBW-2	13.2	0.0
11057A	EUTELSAT 16A	16.0	0.1
19050A	AMOS-17	17.0	0.0
96020A	INMARSAT 3-F1	17.1	6.0
08011A	AMC-14	18.0	20.9
02001A	USA 164 (MILSTAR-2 3)	19.0	8.7
07016A	ASTRA 1L	19.2	0.1
06012A	ASTRA 1KR	19.2	0.0
08057A	ASTRA 1M	19.2	0.0
11041A	ASTRA 1N	19.2	0.0
00054A	ASTRA 2B	19.6	4.4
11049B	ARABSAT-5C	20.1	0.1
12040A	TIANLIAN 1-03	20.4	0.1
13011A	SBIRS GEO-2 (USA 241)	20.6	2.2
14058A	LUCH (OLYMP)	21.5	0.0
12062B	EUTELSAT 21B	21.6	0.1
10021A	ASTRA 3B	23.5	0.0
01025A	ASTRA 2C	23.7	2.7
19014A	WGS 10 (USA 291)	24.0	0.1
13038A	ALPHASAT	24.9	2.4
07056B	SKYNET 5B	25.1	0.1
13044A	EUTELSAT 25B	25.5	0.1
18090A	ES'HAIL 2	25.8	0.0
06051A	BADR-4	26.0	0.1
08034B	BADR-6	26.0	0.1
10025A	BADR-5	26.0	0.0
15065B	BADR-7 (ARABSAT-6B)	26.0	0.0
16052A	USA 270	26.3	0.4
14089A	ASTRA 2G	28.2	0.1
12051A	ASTRA 2F	28.2	0.1
13056A	ASTRA 2E	28.5	0.1
93056A	UFO 2 (USA 95)	28.7	11.3
05005A	XTAR-EUR	29.0	0.0
19021A	ARABSAT-6A	30.4	0.0
10032B	ARABSAT-5A	30.5	0.1
12043B	HYLAS 2	31.0	0.0
19049A	EDRS-C	31.2	0.0
14011B	ASTRA 5B	31.5	0.0
16015A	IRNSS-1F	32.4	2.2
11016A	INTELSAT NEW DAWN	32.8	0.0
97026A	GALAXY 25 (G-25)	32.9	1.8
09008B	EUTELSAT 33E	33.1	0.1
90079A	SKYNET 4C	33.3	14.0
13062A	RADUGA-1M 3	35.0	0.3
19080A	TIBA-1	35.5	0.0
09065A	EUTELSAT 36B	35.9	0.1
15082A	EXPRESS-AMU1	36.1	0.0
14006B	ATHENA-FIDUS	37.8	0.0
11042A	PAKSAT-1R	38.0	0.0
03014A	ASIASAT 4	38.2	0.0
19007A	HELLAS-SAT 4 & SGS-1	39.0	0.1
17040A	HELLAS-SAT 3	39.0	0.1
15012A	EXPRESS-AM7	40.0	0.0
02040B	METEOSAT-8 (MSG-1)	41.4	6.6

14007A	TURKSAT 4A	42.0	0.0
08030B	TURKSAT 3A	42.0	0.1
11077A	NIGCOMSAT 1R	42.5	0.0
03026A	THURAYA-2	44.0	5.9
96021A	ASTRA 1F	44.3	0.1
17046A	COSMOS 2520	45.0	0.0
18074A	AZERSPACE 2 (IS-38)	45.1	0.0
13006B	AZERSPACE 1	46.0	0.0
05041B	SYRACUSE 3A	47.0	0.0
01019A	INTELSAT 10 (IS-10)	47.5	3.8
12016A	YAHSAT 1B	47.6	0.0
07037A	INSAT-4CR	47.9	1.2
17031A	GSAT-19	47.9	0.1
19007B	GSAT-31	47.9	0.1
08065B	EUTELSAT 48D	48.1	0.1
00028A	EUTELSAT 80A	48.1	3.3
19031A	YAMAL 601	49.0	0.0
15060A	TURKSAT 4B	50.0	0.0
97053A	NSS-5	50.5	5.9
16001A	BELINTERSAT-1	51.5	0.1
15023A	TURKMENALEM52E/MONACOSAT	52.0	0.0
11016B	YAHSAT 1A	52.5	0.0
12075A	SKYNET 5D	52.8	0.1
14064A	EXPRESS-AM6	53.0	0.0
12070A	YAMAL 402	54.9	0.0
18089A	GSAT-29	55.0	0.1
13034A	IRNSS-1A	55.0	30.8
18035A	IRNSS-1I	55.1	29.1
11022A	GSAT-8	55.1	0.0
14017A	IRNSS-1B	55.1	28.7
14078A	GSAT-16	55.1	0.1
14010A	EXPRESS-AT1	56.0	0.0
17025A	INMARSAT 5-F4	56.5	0.0
17016A	WGS F9 (USA 275)	56.8	0.0
09058A	NSS-12	57.0	0.0
00081A	ASTRA 2D	57.3	5.7
14023B	KAZSAT-3	58.5	0.0
12008A	BEIDOU 11	58.8	1.4
18110C	TJS-3 AKM	58.9	0.2
04008A	EUTELSAT 7A	59.7	0.2
16053B	INTELSAT 33E (IS-33E)	60.0	0.0
09017A	WGS F2 (USA 204)	60.2	0.0
04007A	ABS-4 (MOBISAT-1)	61.0	0.4
06018A	GOES 13	61.2	0.3
19049B	INTELSAT 39 (IS-39)	62.0	0.0
13073A	INMARSAT 5-F1	62.6	0.0
18105A	GSAT-7A	62.8	0.1
09054B	COMSATBW-1	63.0	0.1
97076A	ASTRA 1G	63.3	4.5
97007A	INTELSAT 26 (IS-26)	63.6	9.0
05044A	INMARSAT 4-F2	63.9	3.0
02041A	INTELSAT 906 (IS-906)	64.1	0.2
00068A	INTELSAT 12 (IS-12)	64.2	2.8
13045A	AMOS-4	65.0	0.1
10065B	INTELSAT 17 (IS-17)	66.0	0.0
11019A	SBIRS GEO-1 (USA 230)	66.2	2.1
16053A	INTELSAT 36 (IS-36)	68.5	0.0
12043A	INTELSAT 20 (IS-20)	68.5	0.0
18107A	COSMOS 2533	70.0	0.0
12069A	EUTELSAT 70B	70.5	0.1
89077A	FLTSATCOM 8 (USA 46)	71.9	13.0
12011A	INTELSAT 22 (IS-22)	72.1	0.0
18100B	GSAT-11	74.0	0.0
16054A	INSAT-3DR	74.0	0.1

13044B	GSAT-7	74.0	0.1
16060A	GSAT-18	74.0	0.1
14001A	GSAT-14	74.1	0.1
16038A	ABS-2A (MONGOLSAT-1)	74.8	0.0
14006A	ABS-2	74.9	0.0
15044A	MUOS-4	75.0	3.0
03057A	UFO 11 (USA 174)	75.2	6.0
15074A	ELEKTRO-L 2	76.0	1.1
12013A	APSTAR 7	76.5	0.0
16072A	TIANLIAN 1-04	77.1	0.3
14002A	THAICOM 6	78.5	0.1
16031A	THAICOM 8	78.5	0.1
18050A	FENGYUN 2H	79.1	0.8
19017A	TIANLIAN 2-01	80.0	2.1
08019A	TIANLIAN 1-01	80.0	4.0
20017A	BEIDOU-3 G2	80.0	3.0
15075A	COSMOS 2513	80.3	0.1
14043A	USA 253	81.0	0.2
15073A	CHINASAT 1C (ZX 1C)	81.5	1.5
02015A	JCSAT-2A	81.9	2.8
13038B	INSAT-3D	82.1	0.1
15041A	GSAT-6	82.3	0.1
11034A	GSAT-12	83.0	0.0
14061A	IRNSS-1C	83.0	1.9
20005A	GSAT-30	83.1	0.1
12051B	GSAT-10	83.1	0.1
19070A	TJS-4	83.5	0.1
12059A	BEIDOU 16	84.0	1.8
95035B	TDRS 7	84.8	14.2
07063B	HORIZONS 2	84.9	0.0
10002A	RADUGA-1M 2	85.0	0.0
09067A	INTELSAT 15 (IS-15)	85.2	0.0
07007A	INSAT-4B	85.5	0.3
08066A	FENGYUN 2E	86.2	4.2
11035B	KAZSAT-2	86.5	0.0
04004A	USA 176 (DSP 22)	87.4	9.1
12067A	CHINASAT 12 (ZX 12)	87.5	0.0
11022B	ST-2	88.0	0.0
12003A	WGS F4 (USA 233)	88.4	0.0
00034A	TDRS 8	88.9	9.2
14082A	YAMAL 401	90.0	0.0
09032A	MEASAT-3A	91.5	0.0
06056A	MEASAT-3	91.5	0.1
14054B	MEASAT-3B	91.5	0.0
08028A	CHINASAT 9 (ZX 9)	92.2	0.0
15019A	BEIDOU 17	92.4	53.1
17040B	GSAT-17	93.5	0.1
15065A	GSAT-15	93.5	0.1
11038A	BEIDOU 9	94.6	54.4
14023A	LUCH 5V	94.7	0.6
18049A	SES-12	95.0	0.0
13071A	SES-8	95.0	0.0
07007B	SKYNET 5A	95.2	0.4
11073A	BEIDOU 10	95.4	51.7
15053A	BEIDOU 20	96.0	52.7
16021A	BEIDOU IGSO-6	96.2	57.0
08003A	EXPRESS-AM33	96.5	0.0
17024A	GSAT-9	97.3	0.1
13020A	CHINASAT 11 (ZX 11)	98.0	0.0
12028A	CHINASAT 2A (ZX 2A)	98.3	0.0
08001A	THURAYA-3	98.5	3.8
14090A	FENGYUN 2G	99.5	2.0
98050A	ASTRA 2A	100.0	1.5
09042A	ASIASAT 5	100.5	0.0

16048A	TIANTONG-1 1	101.4	3.1
17035A	CHINASAT 9A (ZX 9A)	101.4	0.0
05023A	EXPRESS-AM3	103.0	1.7
15063A	CHINASAT 2C (ZX 2C)	103.4	0.0
95038A	USA 113	103.9	11.4
19073A	BEIDOU-3 IGSO-3	104.7	58.5
16077A	FENGYUN 4A	104.7	0.2
00016A	ASIASTAR	105.0	3.9
11069A	ASIASAT 7	105.5	0.0
15083A	GAOFEN 4	105.7	0.1
19035A	BEIDOU-3 IGSO-2	106.0	55.1
17001A	TJS-2	107.5	0.0
18064A	TELKOM 4 (MERAH PUTIH)	108.0	0.0
09027A	SES-7 (PROTOSTAR 2)	108.3	0.1
16013A	SES-9	108.3	0.0
16052B	USA 271	109.2	0.1
17059B	BSAT-4A	109.8	0.1
07036B	BSAT-3A	109.9	0.0
10056B	BSAT-3B	109.9	0.0
11041B	BSAT-3C (JCSAT-110R)	110.0	0.0
16082A	JCSAT-15	110.1	0.0
11026A	CHINASAT 10 (ZX 10)	110.5	0.1
17018A	CHINASAT 16 (SJ-13)	110.5	0.1
16037A	BEIDOU-2 G7	110.6	1.4
13050A	AEHF-3 (USA 246)	111.0	0.7
16003A	IRNSS-1E	111.7	29.1
15018A	IRNSS-1D	112.0	29.0
19023A	BEIDOU-3 IGSO-1	112.0	55.4
12002A	FENGYUN 2F	112.1	1.9
18057A	BEIDOU IGSO-7	112.8	55.0
09046A	PALAPA D	113.0	0.1
06034A	KOREASAT 5 (MUGUNGWHA 5)	113.1	0.0
17067A	KOREASAT 5A	113.2	0.0
07031A	ZHONGXING-6B	115.5	0.0
17023A	KOREASAT 7	115.9	0.0
10070B	KOREASAT 6	116.0	0.0
99046A	ABS-7	116.1	0.8
17007A	TELKOM 3S	118.0	0.0
18044A	BANGABANDHUSAT-1	119.1	0.0
10036A	BEIDOU 5	119.3	54.2
05028A	THAICOM 4	119.5	0.0
10068A	BEIDOU 7	119.8	51.6
14052A	ASIASAT 6	119.9	0.0
11013A	BEIDOU 8	120.2	59.2
17057A	ASIASAT 9	122.1	0.0
16050A	JCSAT-16	124.0	0.1
12023A	JCSAT-13	124.0	0.0
10042A	CHINASAT 6A (ZX 6A)	125.1	0.1
19097A	SHIJIAN-20 (SJ-20)	125.4	0.1
10064A	ZHONGXING-20A	126.4	0.7
17048A	QZS-3 (MICHIBIKI-3)	127.0	0.1
06033A	JCSAT-3A	128.0	0.0
18037A	COSMOS 2526	128.1	0.0
10032A	COMS 1	128.2	0.0
20013B	GEO-KOMPSAT-2B	128.3	0.0
18100A	GEO-KOMPSAT-2A	128.3	0.0
15067A	LAOSAT 1	128.5	0.0
16027A	IRNSS-1G	129.6	2.3
11047A	CHINASAT 1A (ZX 1A)	129.9	0.1
19012A	CHINASAT 6C (ZX 6C)	130.0	0.1
19001A	CHINASAT 2D (ZX 2D)	130.0	0.0
12023B	VINASAT-2	131.9	0.0
08018A	VINASAT-1	132.0	0.0
06010A	JCSAT-5A	132.1	0.0

18041A	APSTAR 6C	134.0	0.1
18036A	USA 283	134.1	0.0
00065A	USA 153	134.5	7.8
20013A	JCSAT-17	136.0	7.0
99006A	JCSAT-4A	136.0	4.5
02035B	N-STAR C	136.0	6.6
01009A	USA 157 (MILSTAR-2 2)	137.7	9.3
10045A	QZS-1 (MICHIBIKI-1)	138.0	41.6
18069A	TELSTAR 18V (APSTAR 5C)	138.0	0.0
17028A	QZS-2 (MICHIBIKI-2)	138.3	43.1
18009A	SBIRS GEO-3 (USA 282)	139.0	5.1
17062A	QZS-4 (MICHIBIKI-4)	139.5	40.8
14010B	EXPRESS-AT2	139.9	0.0
13077A	EXPRESS-AM5	140.0	0.0
18085A	BEIDOU-3 G1	140.1	1.9
15054A	SKY MUSTER (NBN1A)	140.3	0.0
10001A	BEIDOU 3	140.4	1.4
14060A	HIMAWARI-8	140.7	0.0
16064A	HIMAWARI-9	140.8	0.0
15059A	APSTAR 9	142.0	0.0
05009A	INMARSAT 4-F1	143.5	3.2
08038A	SUPERBIRD-C2	144.0	0.0
19027A	BEIDOU-2 G8	144.5	1.1
16060B	SKY MUSTER 2	144.8	0.0
06004A	HIMAWARI-7 (MTSAT-2)	145.0	0.0
02062A	NIMIQ 2	145.5	4.2
19009A	NUSANTARA SATU	146.0	0.0
99013A	ASIASAT 3S	147.5	4.7
16075A	WGS F8 (USA 272)	149.8	0.0
19091A	JCSAT-18 (KACIFIC 1)	150.0	0.0
16039A	BRISAT	150.5	0.0
07044A	OPTUS D2	152.0	0.0
16028A	JCSAT-2B	154.0	0.0
15046A	TJS-1	155.0	0.1
03028B	OPTUS C1	155.7	1.3
14054A	OPTUS 10	156.0	0.1
09044B	OPTUS D3	156.0	0.1
05046A	TELKOM 2	157.0	0.0
00072A	INTELSAT 1R (IS-1R)	157.1	2.6
99053A	ABS-6	159.0	0.0
06043B	OPTUS D1	160.0	0.0
10057A	BEIDOU 6	160.0	0.8
18033A	SUPERBIRD 8	162.0	0.0
05012A	APSTAR 6	162.9	1.5
16065A	SHIJIAN-17 (SJ-17)	163.2	0.1
03053A	YAMAL 202	163.5	0.7
19095A	ELEKTRO-L 3	165.8	0.3
12030A	INTELSAT 19 (IS-19)	166.0	0.0
11074B	LUCH 5A	166.9	3.1
09044A	JCSAT-RA (JCSAT-12)	168.9	0.0
18074B	HORIZONS 3E	169.0	0.0
11032A	TIANLIAN 1-02	171.1	2.8
98016A	UFO 8 (USA 138)	171.6	8.6
17029B	EUTELSAT 172B	172.0	0.0
95003A	UFO 4 (USA 108)	172.2	9.9
18110A	TJS-3	173.0	0.1
05052A	EUTELSAT 174A	174.0	0.1
15036A	WGS F7 (USA 263)	175.0	0.0
00059A	NSS-11 (AAP-1)	176.0	0.1
96070A	INMARSAT 3-F3	178.1	5.5
15042A	INMARSAT 5-F3	179.6	0.0

CHAPTER FOUR: MULTIPLE-ACCESS TECHNIQUES IN COMMUNICATION

A single satellite transponder allows simultaneous communications among multiple users. This is also the case with other communication channels like radio, fiber optic cables and microwave channels. The technique of using one channel to establish multiple communication links is called multiplexing. Note the difference between a communication channel and a communication link. A channel can be a single line of cable or any physical medium (such as antenna to antenna) between two parties. A link is a communication line between one user and another. A channel can contain thousands of links such that a single fiber optics cable can link so many subscribers in a GSM telephone system. Individual links are not separate physical structures like channels. There are 3 major multiplexing methods in multimedia communication.

4.1 FDMA

FDMA stands for Frequency Division Multiple Access. All the early satellites used this technique. The communication space is divided into several frequency channels that serve as individual links just like FM radio stations. A subscriber is allocated a link for each communication session which is recovered after the session is ended. Note that no matter what frequency channel the link may belong to, the same antennas and the same set of processing equipment are used. Getting a good and uniform power amplification for this broad frequency range can be a challenge.

4.2 TDMA

TDMA stands for Time Division Multiple Access. In TDMA all communication links use the same carrier frequency. They are allocated time frames in which the channel handles their traffic bursts.

Advantages of TDMA over FDMA

1. Each channel operates at a single carrier frequency. This results in a better power amplification for the signal.

2. For satellite communication, the earth stations in FDMA must transmit and receive at multiple frequencies and the required frequency selectivity is expensive to accomplish. In TDMA there is only one frequency.
3. TDMA is ideally suited for digital communication. This is not the case with FDMA.
4. TDMA is ideally suited to a common practice known as demand assignment where a subscriber is allocated a share of the channel time exactly in line with his demand. FDMA cannot allocate frequencies this way.

4.3 CDMA

CDMA stands for Code Division Multiple Access. In this technique only one carrier frequency is used. Each communication link uses a different binary sequence to modulate the carrier. It is a direct sequence spread spectrum technique. The frequency spectrum is however higher than the case with TDMA and FDMA. CDMA offers the advantage of superior noise rejection and is capable of accommodating much higher number of links.

4.4 DAMA

Demand Assigned Multiple Access (DAMA) is a technology used to assign a channel to clients that don't need to use it constantly. DAMA systems assign communication channels based on requests issued from user terminal to a network control system. When the circuit is no longer in use, the channels are then returned to the central pool for reassignment to other users.

Channels are typically a pair of carrier frequencies (one for transmit and one for receive), but can be other fixed bandwidth resources such as timeslots in a TDMA burst plan or even physical party line channels. Once a channel is allocated to a given pair of nodes, it is not available to other users in the network until their session is finished.

It allows utilizing of one channel (radio or baseband frequency, timeslot, etc.) by many users sequentially *at different times*. This technology is mainly useful with sparsely used networks of transient clients, as opposed to *PAMA (Permanently Assigned Multiple Access)*. By using DAMA technology the number of separate nodes that can use a limited pool of circuits can be greatly increased at the expense of no longer being able to provide simultaneous access for all possible pairs of nodes. A five-channel DAMA network can only have five simultaneous conversations but could have any number of nodes. A five-channel PAMA network permanently supports five simultaneous conversations, with channel ownership remaining with their permanently assigned nodes even when idle.

DAMA and PAMA are related only to channel/resource *allocation* and should not be confused with the Multiple access/multiplexing methods (such as FDMA frequencies, TDMA slots, CDMA codes, or others) intended to divide a single communication channel into multiple *virtual* channels. These systems typically use resource allocation protocols that allow a more rapid (although often less deterministic, consider CDMA collisions) near-real-time allocation of bandwidth based on demand and data priority. However, in sparsely allocated multiple-access channels, DAMA *can* be used to allocate the individual *virtual channel* resources provided by the multiple-access channel. This is most common in environments that are sufficiently sparsely utilized that there is no need to add complexity just to recover "conversation gap" idle periods.

DAMA is widely used in satellite communications, especially in VSAT systems. It is very effective in environments comprising multiple users each having a low to moderate usage profile.

DAMA is often used in military environments due to the relative simplicity of implementation, ease of modeling, and the fact that military usage profiles are a very good fit. In military SATCOM, it has the added advantage that it can function in a bent pipe environment, thus requires no special security or coordination hardware on the satellite. This allows the master and slave ground stations to be upgraded repeatedly to change or improve security and compression without requiring an expensive satellite replacement

CHAPTER FIVE: TELEVISION

A television set works by displaying the output of its video signal across the screen in hundreds of lines from left to right and from top to the bottom just as you would read a page of this document in many lines from left to right and from top to bottom. The following terms are commonly used in television terminology and must be well understood.

5.1 TELEVISION TERMINOLOGIES

TELEVISION FRAME

Frame is to television the same as page is to a book. The information visible at any instance of time on a television screen is a frame of information just as the information available on a page of a book.

FRAME RATE

Frame rate is the number of times a complete frame is displayed per second. In the television standard known as NTSC which is operational in the United States of America, there are 525 lines per frame. Each of these lines is traced out 30 times per second. The frame rate is therefore 30 in NTSC. In Nigeria and most of Europe, the operational standard is PAL. There are 625 lines per frame and the frame rate is 25.

INTERLACED SCANNING

If the 625 lines in PAL system are displayed sequentially the television beam will travel from top to bottom 25 times every second. The picture on the screen will be flickering. To prevent flickering, interlaced scanning is used. The 625 lines will not be traced out sequentially. The beam will pass from pass from top to bottom tracing out only odd numbered lines. Then it will go back to the top and begin to trace out even numbered lines. Even though the frame rate is 25, the beam will travel from top to bottom 50 times per second. The effect of persistence will make the picture to appear continuous on the screen.

PERSISTENCE

A television does not display information on a screen in a kind of shower. Rather the information comes in form of a tiny beam of light that moves from left to right and from top to bottom to trace out an image. In order to show a complete image, what is displayed at the top must still remain in our sight even when the beam is now at the bottom. The length of time an image stays on the screen after the beam displaying it has been removed is called persistence. There are two sources of persistence and both sources add up to the final effect. The first is phosphor persistence and the second is persistence of vision. Phosphor is the substance used to cover the entire screen. When a beam of electrons strikes phosphor it produces a visible glow that depends on the intensity of the electron beam. This glow persists for fractions of a second after the electron beam has been removed. Similar effects are also observed in display systems that do not use electron beams such as Liquid Crystal Display (LCD) and Light Emitting Diode (LED) display.

Persistence of vision is the way human eyes perceives an image as still being present some fractions of a second after the image has been removed. If you flash a light on a screen about 5 times every second, the human eyes will notice that the light is not steady but if you flash it 100 times every second, the eyes will think that the light is permanently on the screen because of the persistence of human vision.

PROGRESSIVE SCANNING

The more times you scan a signal per second, the more data you generate thereby increasing the required bandwidth. Interlaced scanning solves this problem by scanning at 25 frames per second and displaying at 50 frames per second. In situations where the issue of bandwidth is not critical, progressive scanning is used. All the lines are displayed sequentially from line 1 to line 2, all the way to line 625. The frame rate must be high in this case so that there will be no flicker. In PAL, a frame rate of 50 is used for progressive scanning.

SYNCHRONIZATION

Every television video signal must have a horizontal and vertical synchronization signal or horizontal sync and vertical sync superimposed on it. This is necessary so that the television can tell the display information for line 1 different from that of line 2. The horizontal trace is in form of a pulse that blanks out the display for a short duration while the electron beam returns to the beginning of line 2 after completing line 1. This time interval is called the horizontal retrace interval.

When all the horizontal lines have been displayed, the electron beam moves from the lower right hand corner of the screen (end of line 625) to the upper left hand

corner (start of line 1) for the next frame. The time this movement takes is called the vertical retrace interval. The duration of the vertical retrace interval is much higher than that of the horizontal retrace interval. This long duration allows for a longer travel by the beam to the beginning of next frame.

PIXEL

Pixel literally means picture element. It is the smallest area that carries video information. The entire frame is divided into pixels. The total number of pixels determines the resolution of the video signal which is often expressed as the product of two numbers like 1024x768. The first number is the number of pixels from left to right while the second figure is the number of pixels from top to bottom. A high resolution or high definition video will have the pixels closely packed together giving rise to more pixels per unit area than that of a low resolution video. High resolution videos require higher bandwidth for transmission.

ASPECT RATIO

The ratio of the width to the height of the frame is called the aspect ratio. In analog television, it is 4:3 while in digital video standards, it is 16:9.

AUDIO SIGNALS IN TELEVISION

The audio signal is captured and processed separately from the video signal. It is a much simpler signal with a much lower bandwidth and no synchronization requirement. It is added to the video signal before transmission and recovered from the video signal after reception for a separate processing before being connected to the speaker of the television.

5.2 HDTV

HDTV stands for High Definition Television. The term has had various standards in the history of television but the current definition of HDTV is that of a digital television whose resolution is superior to that of an analog television while maintaining an aspect ratio of 16:9.

HDTV CODING

To convert the analog video to a digital format, ITU-R 601 (also called rec.601) 4:2:2 format is used. A video signal is composed of green, red and blue components. In addition to providing green colour information, the green channel provides the illumination information. Luminance is the black-and-white detail. This is termed Y. the red colour information is termed R-Y while the blue colour information is B-Y. The three signals generated are Y, R-Y and B-Y. The sampling rate used for digitization is in the ratio of 4:2:2 respectively. The actual sampling frequencies used are 13.5MHz : 6.75MHz : 6.75MHz. The samples are digitized into 10-bit words per sample. The total number of bits generated per second is 135Mb/s + 67.5Mb/s + 67.5Mb/s. These three signals are then multiplexed together using TDMA to obtain a serial data bit rate of 270Mb/s. but a television channel only has a bandwidth of 6MHz. The video signal must therefore be compressed using standard compression technique called MPEG2, The output of the compression process is then multiplexed with the audio signal and processed for transmission. This is how HDTV signal is generated.

5.3 VIDEO SPECIFICATIONS

A specification of 1920 X 1080i means that the first number 1920 is the number of pixels per line while 1080 means there are 1080 lines. The letter i means interlaced scanning. If the specification is stated as 1920 X 1080p, the display uses progressive scanning. Computer monitors use progressive scanning. Sometimes 1080p is stated instead of 1920 X 1080p. Abbreviating to 1080p is possible because the aspect ratio is always 16:9 in all HDTV, if 1080 is stated then the first figure is calculated as $1080 \times 16/9 = 1920$. Sometimes the frame rate is also specified. The complete specification can be 1920 X 1080i25 where 25 is the frame rate.

CHAPTER SIX: GSM AND CDMA PHONE SYSTEMS

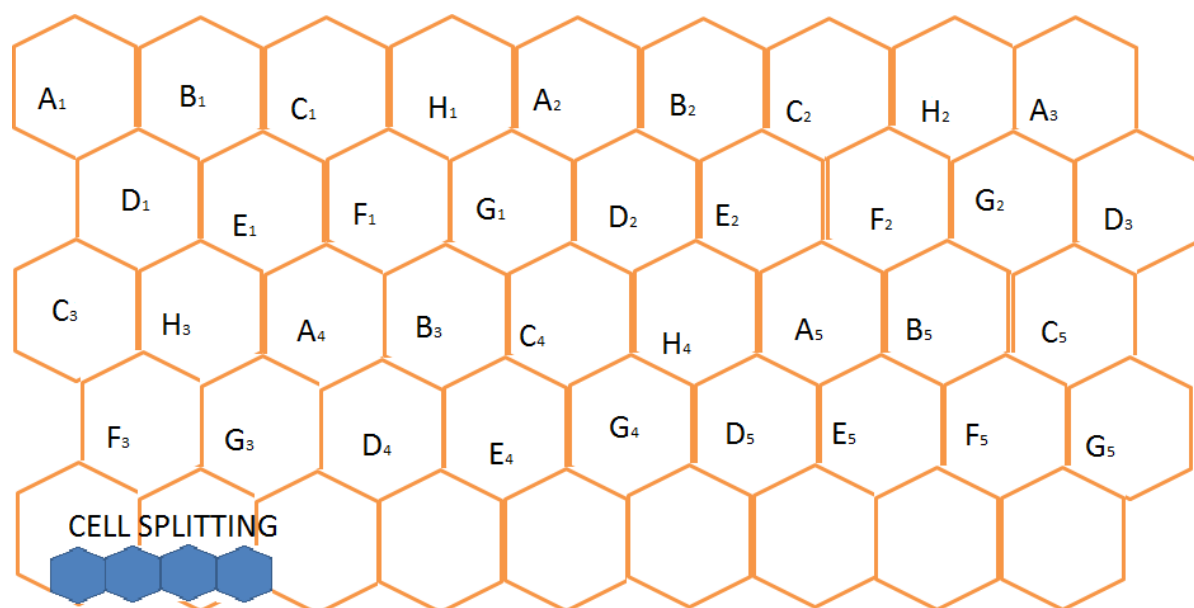
In recent years, GSM phone systems have gained such popularity in Nigeria that it has acquired a subscriber base greater than all other telephone systems put together. It is perhaps the commonest channel for multimedia communication in the country. On a worldwide bases, GSM is in direct competition with CDMA for mobile communication. In the United States of America CDMA dominates while in Europe GSM dominates. The two technologies have a lot in common and share a common history.

6.1 MOBILE PHONE CONCEPTS

Both CDMA and GSM phone systems are called cellular telephone services. The coverage area in each case is divided into cells, each cell is a regular hexagon. This way, every location belongs to a cell. The two major concepts are cell splitting and frequency reuse. Both concepts have led to a high degree of scalability in the deployment of cellular telephone services.

FREQUENCY REUSE

Frequency reuse is the process of using the same carrier frequency in different cells that are geographically separated.



In the above cells A1, A2, A3, A4 et c. use exactly the same carrier frequency, same goes for the cells labeled B1, B2, B3 et c. Such cells are separated as far apart as possible to ensure there is no interference. The system reduces the transmission power to the barest minimum that ensures good reception without causing interference. This way a total of 8 frequency channels can be shared among thousands of cells covering the entire geographical area.

CELL SPLITTING

Cell splitting becomes necessary when the number of subscribers within the existing cells has increased beyond the capacity of the base station. The cells in the congested area are split into more cells and allocated different frequency bands. New base stations are set up. The furthest distance to the nearest base station is thereby reduced. This permits lower transmission power which prevents interference.

6.2 COMPONENTS OF GSM AND CDMA TELEPHONE SYSTEMS

The equipment used in GSM are in 4 locations: the handset, the base station, the base station controller (BSC) and the mobile telephone switching office (MTSO). In CDMA systems BSC is absent. Most of its functions are handled by the base station for which reason CDMA base stations are much more costly to set up.

MOBILE TELEPHONE SWITCHING OFFICE (MTSO)

MTSO contains the switching equipment that connects a caller to other subscribers on the same network; subscribers on other mobile networks and subscribers on land lines. It also contains the database where subscriber information resides. Routing of calls takes place at the MTSO. Billing and authentication are all handled by the MTSO. It is the heart of the mobile telephone system whether GSM or CDMA. It uses a command and control data channel to control the actions of the handsets.

BASE STATION CONTROLLER (BSC)

A BSC can have up to a hundred base stations under its control. It allocates radio channels to the handsets and receives measurements from them. It also handles handover of a moving subscriber from base station to base station. It acts as a traffic

concentrator that channels many low capacity connections into a more efficient but smaller number of connections to the MTSO. The data for the allocation of cells to base stations as well as the cell boundaries and the frequency channel for each cell are contained in the BSC. Another function of the BSC is transcoding. It carries out a translation between the coding used on the mobile network and the coding required on the global telephone circuit.

However CDMA systems do not have BSC. Their functions are carried out by the base stations mostly and also handled at the MTSO.

BASE TRANSCEIVER STATION (BTS) OR BASE STATION

The base station facilitates the wireless communication between the mobile phones and the network. It handles both transmission and reception of signals. Signal power amplification is a base station function. It contains the antenna for sending and receiving signals. The antenna can be omnidirectional or sectorial. A sectorial is a composite antenna made from three sectors set at 120 degrees to one another.

6.3 HOW GSM WORKS

If you find yourself within the coverage area of a GSM telephone system, you are located within a cell. Each cell is allocated a frequency band of 200KHz for transmission in the range of 890MHz to 915MHz. the same cell is also allocated a frequency band of 200KHz for reception in the range 935MHz to 960MHz. The 200KHz range is shared in time slots using TDMA multiple access technique. The time slots are for voice, messages and data. When a handset is switched on, it listens on a preprogrammed channel for signal from a base station. This starts a sequence of communications that identifies the mobile phone to the network; measures its distance to the nearest base stations; picks the optimum base station for the particular handset and determines the power level for its transmission. Registration with MTSO ensures that any call to the handset is forwarded to that set. When a call comes to a handset, the BSC allocates a time slot for the call. If it is unable to find a time slot, the caller will be notified that the subscriber is not reachable.